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Miranda et al.

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(54) **SYSTEMS AND METHOD FOR CAPTURING, PROCESSING, AND DISPLAYING A 360° VIDEO**

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H04N 13/351 (2018.01)
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H04N 23/698 (2023.01)

(52) **U.S. Cl.**
CPC **H04N 13/351** (2018.05); **H04N 13/156** (2018.05); **H04N 23/698** (2023.01)

(58) **Field of Classification Search**

CPC H04N 13/156; H04N 13/351; H04N 21/21805; H04N 21/234345; H04N 21/816; H04N 21/85406; H04N 23/698

See application file for complete search history.

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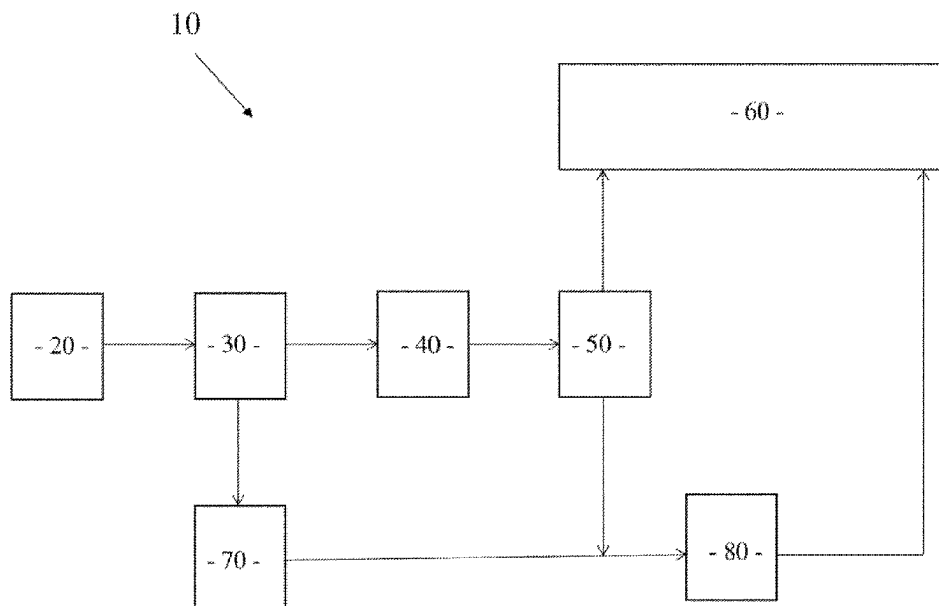
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Primary Examiner — Jessica M Prince

(57) **ABSTRACT**

A 360° video capture, processing and display system includes a 360° video camera to capture a 360° raw video file, and a 360° video processing assembly to generate a 360° equirectangular geographic projection from each frame of the 360° raw video file. A 360° video processing algorithm captures a plurality of viewport images from each of a plurality of equirectangular geographic projections. A composite video displays includes a plurality of viewport images from a corresponding one of each of the 360° equirectangular geographic projections, wherein at least one of a top or bottom composite video display comprises a seamless integration of corresponding ones of the plurality of viewport images. The 360° video capture, processing and display system generates a multi-platform compatible processed 360° video file comprising the plurality of composite video displays, wherein the multi-platform compatible processed 360° video file is viewable on a video display assembly.

18 Claims, 22 Drawing Sheets



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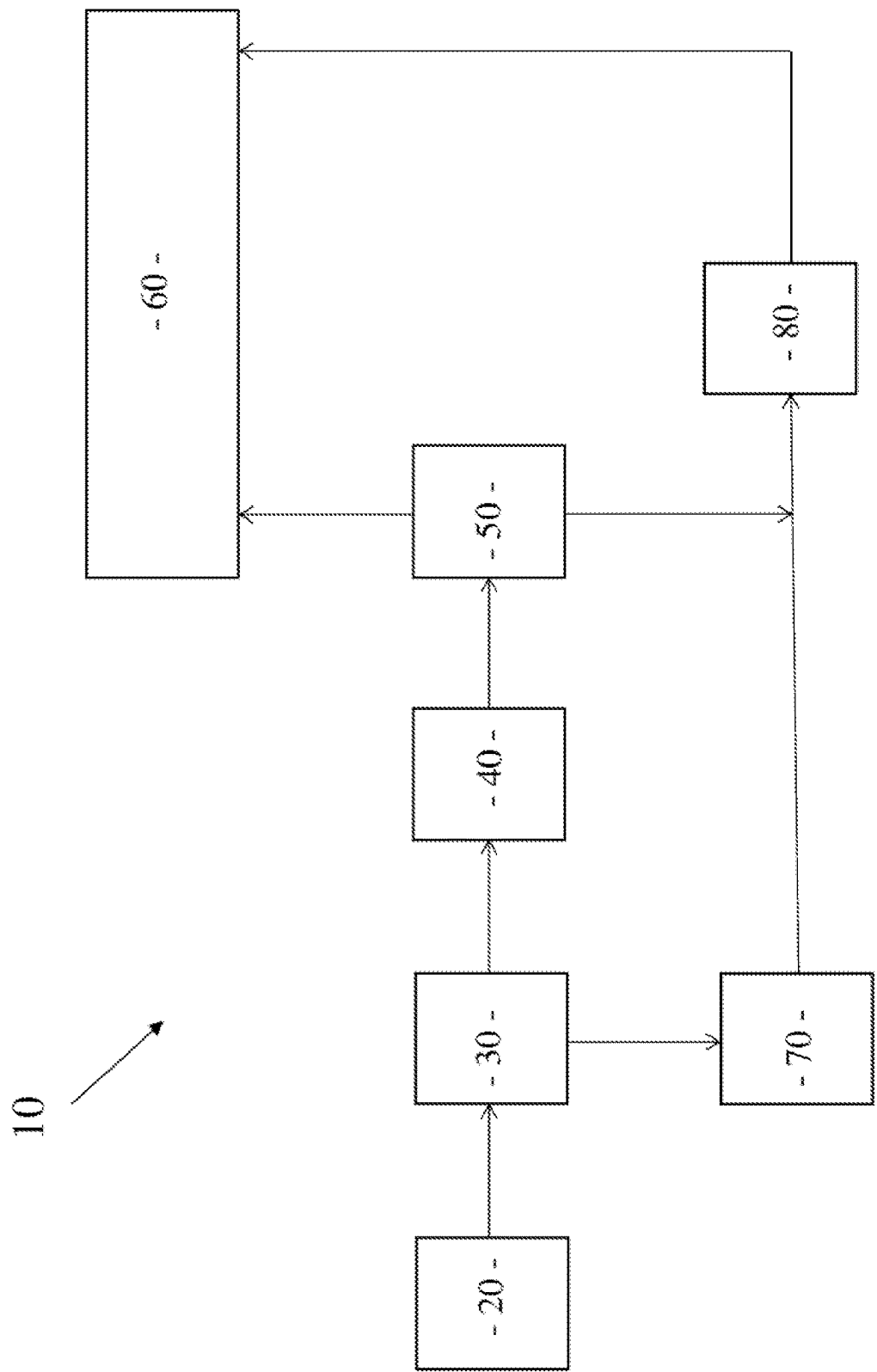


FIG. 1

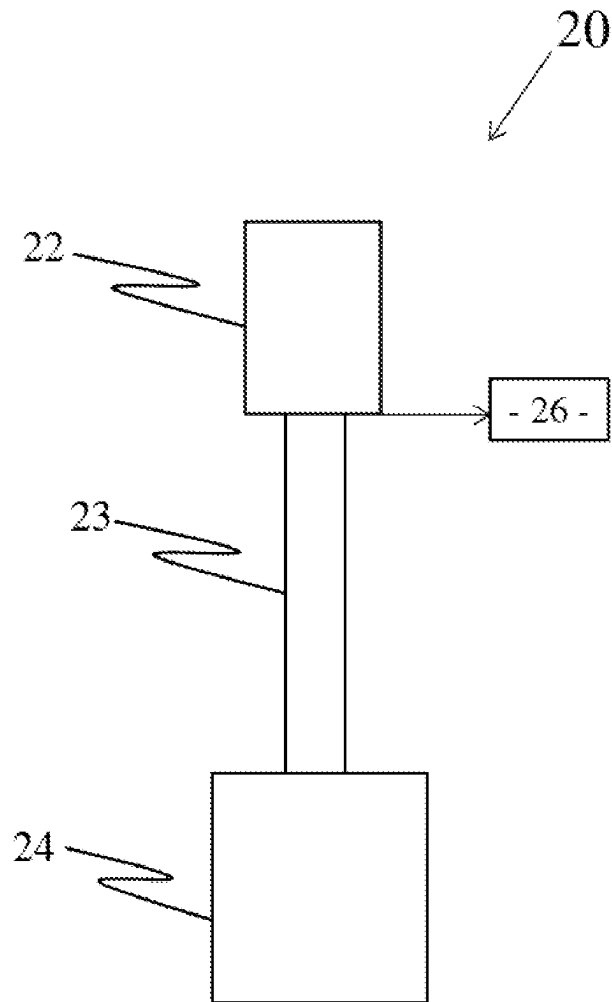
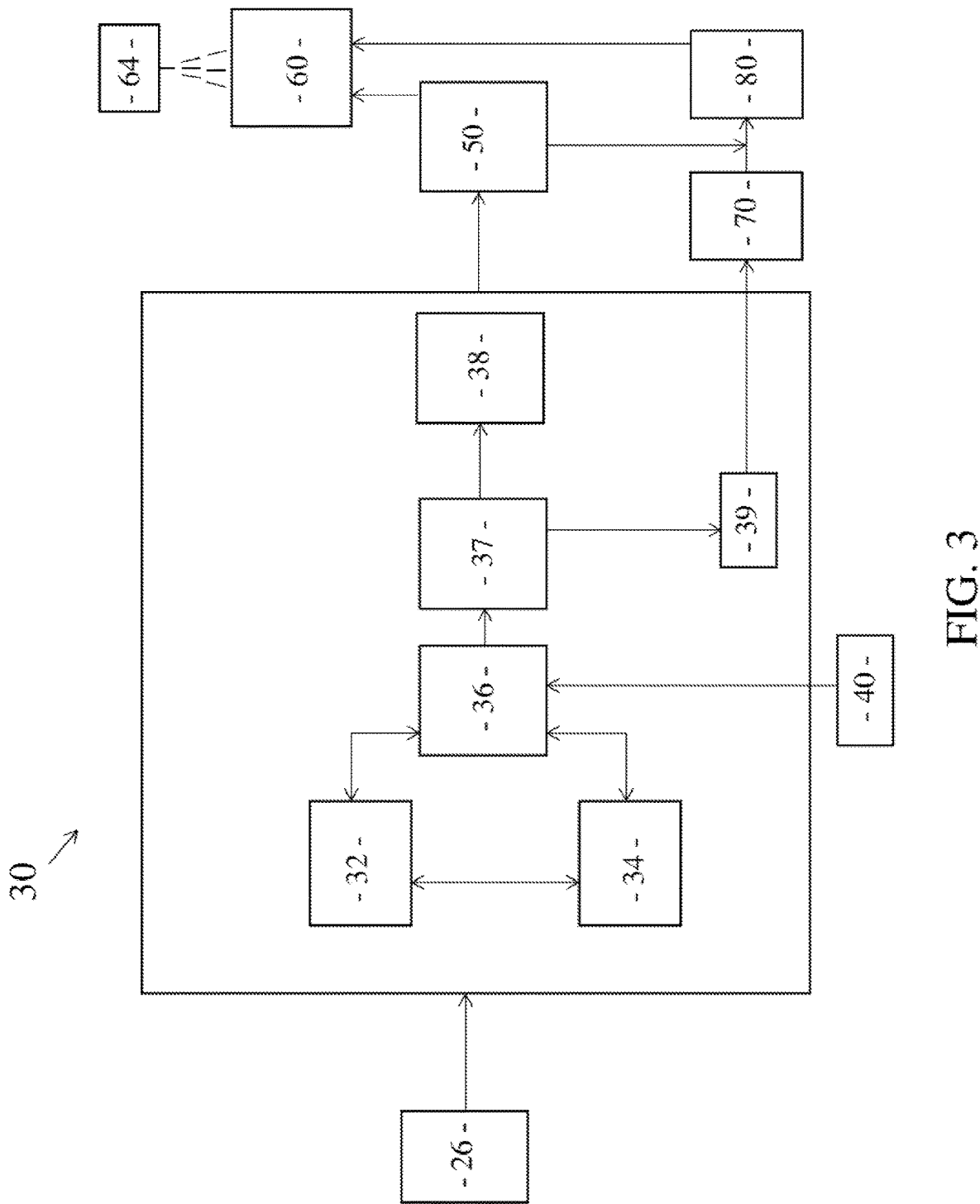


FIG. 2



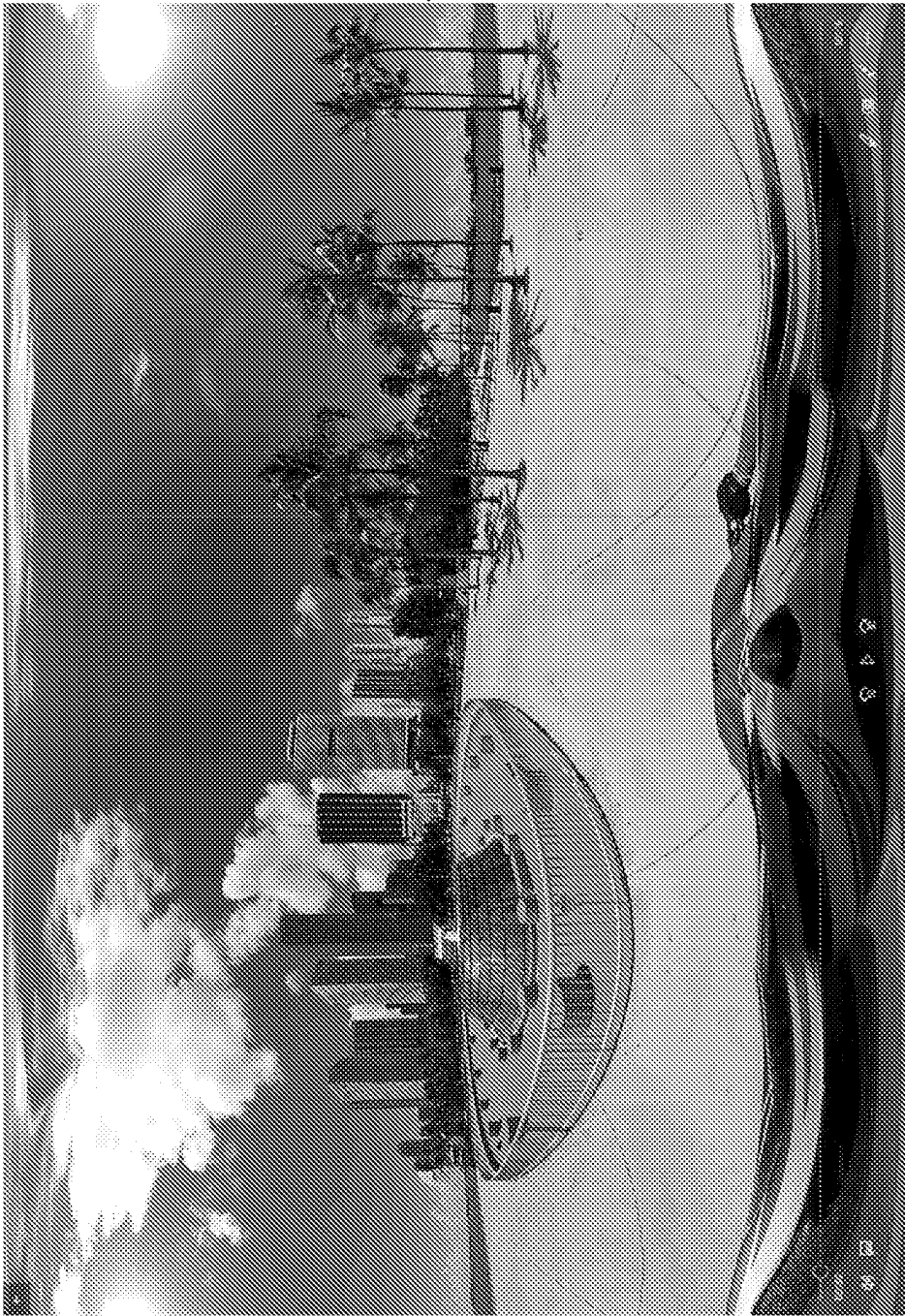


FIG. 4

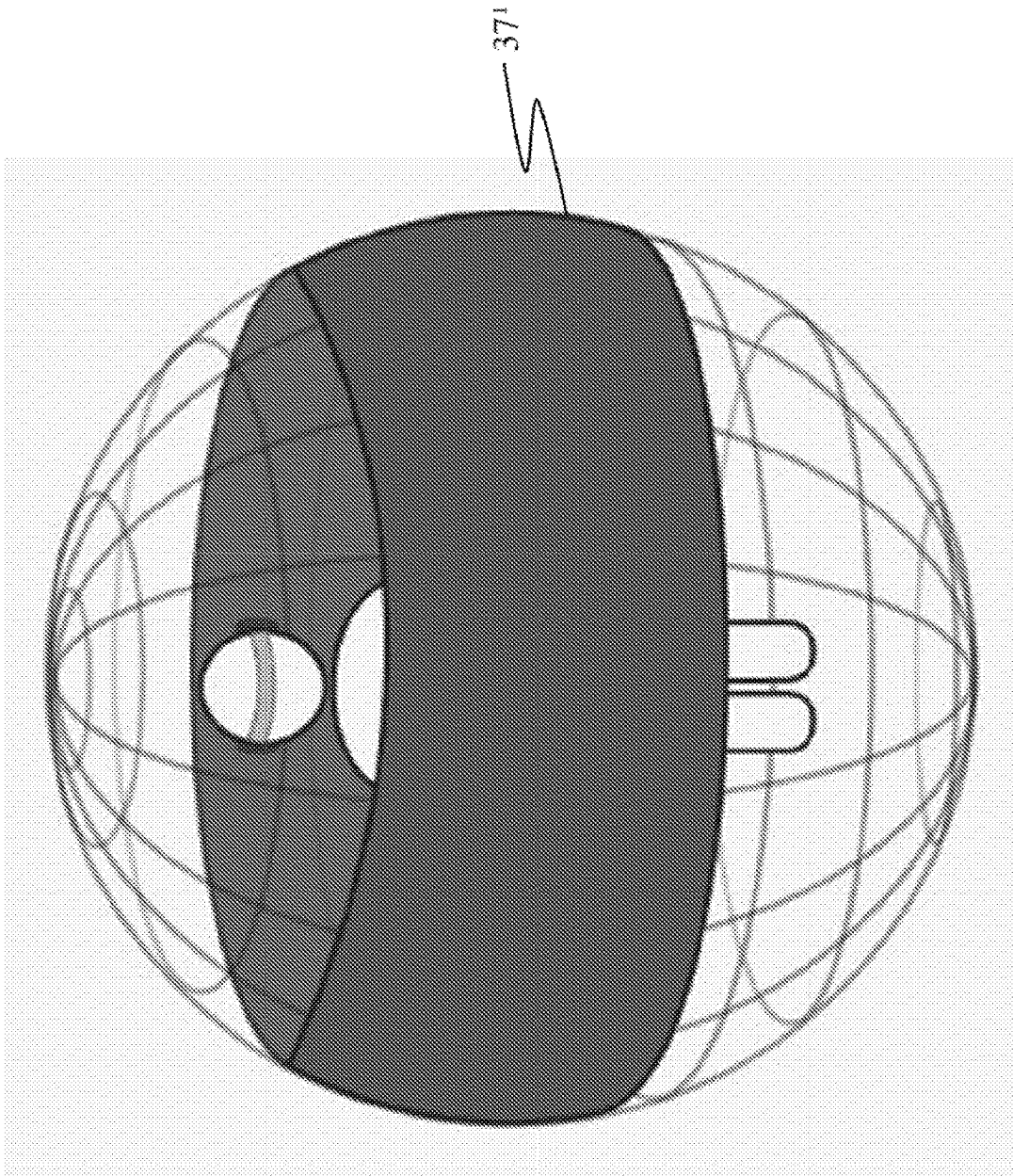


FIG. 5

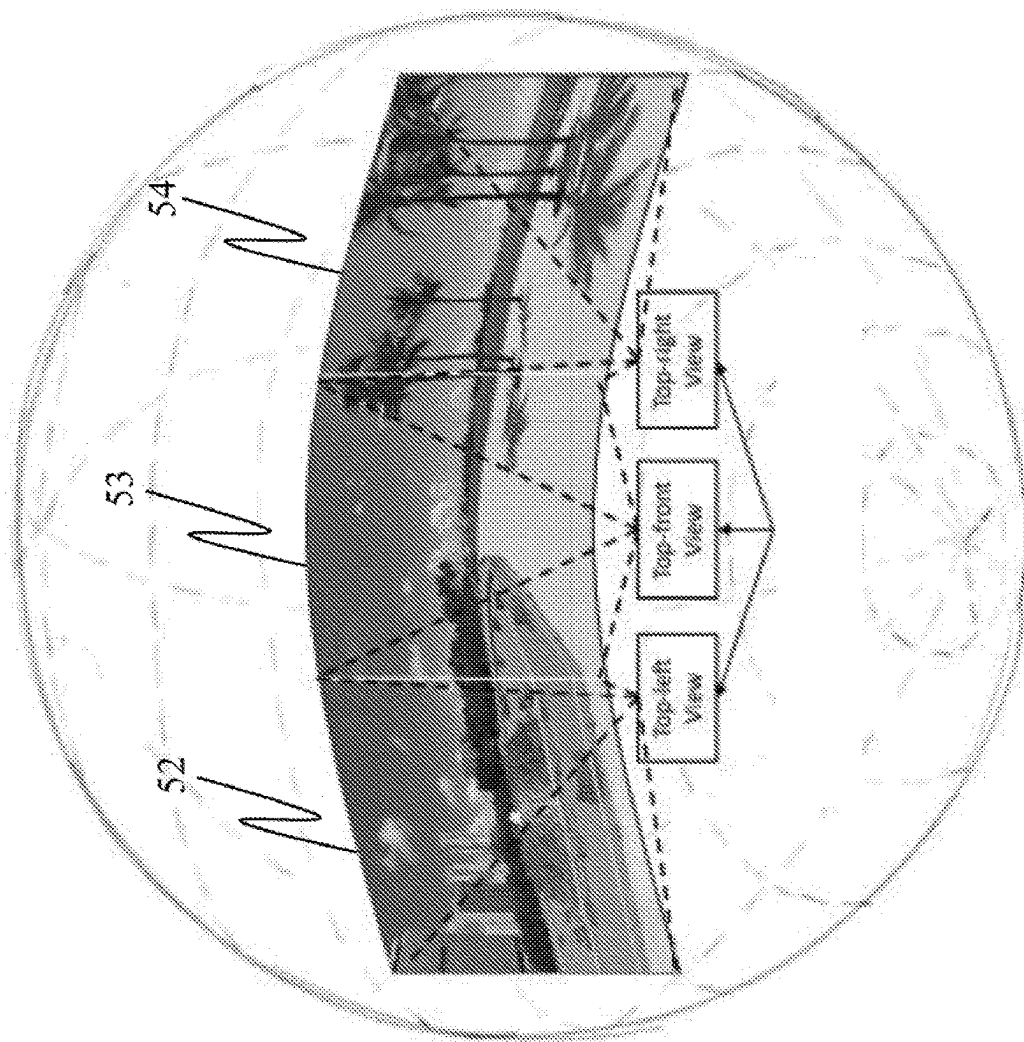


FIG. 6

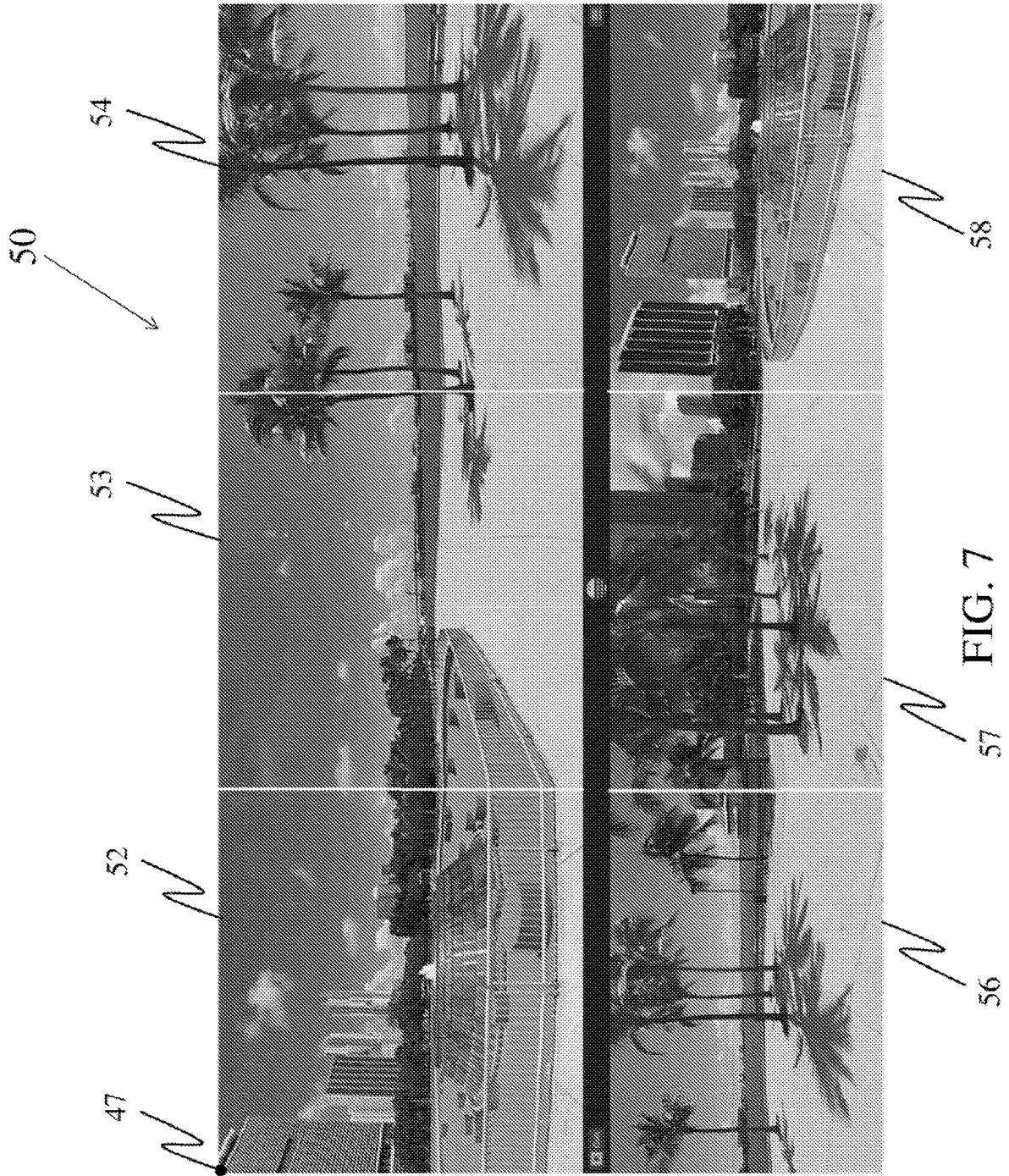


FIG. 7

TABLE 3
Predetermined Video Display Layout (43) - BVT Layout Type 1

Top Viewpoint Height (45):	755	Top Display Height Ratio (44): 58.98%
Navigation Menu (59) Height:	40	Navigation Menu (59) Height Ratio: 3.13%
Bottom Viewpoint Height (45):	485	Bottom Display Height Ratio (44): 37.89%

Top Composite Video Display (51)						
Viewport Image	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
	x (pixels)	y (pixels)				
Top Left (52)	Top_Left_x	Top_Left_y	Top_Left_Width	Top_Left_Height	Top_Left_FOV	Top_Left_Heading_Ang
	0	0	640	755	70	19.7
Top Center (53)	Top_Center_x	Top_Center_y	Top_Center_Width	Top_Center_Height	Top_Center_FOV	Top_Center_Heading_Ang
	640	0	640	755	70	90.0
Top Right (54)	Top_Right_x	Top_Right_y	Top_Right_Width	Top_Right_Height	Top_Right_FOV	Top_Right_Heading_Ang
	1280	0	640	755	70	160.4

Bottom Composite Video Display (55)						
Viewport Image	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
	x (pixels)	y (pixels)				
Bottom Left (56)	Bottom_Left_x	Bottom_Left_y	Bottom_Left_Width	Bottom_Left_Height	Bottom_Left_FOV	Bottom_Left_Heading_Ang
	0	795	640	485	70	179.3
Bottom Center (57)	Bottom_Center_x	Bottom_Center_y	Bottom_Center_Width	Bottom_Center_Height	Bottom_Center_FOV	Bottom_Center_Heading_Ang
	640	795	640	485	70	270.0
Bottom Right (58)	Bottom_Right_x	Bottom_Right_y	Bottom_Right_Width	Bottom_Right_Height	Bottom_Right_FOV	Bottom_Right_Heading_Ang
	1280	795	640	485	70	360.8

FIG. 8

TABLE 4

Predetermined Video Display Layout (43) - BVT Layout Type 2

Top Viewpoint Height (45):	853	Top Display Height Ratio (44):	66.64%
Navigation Menu (59) Height:	40	Navigation Menu (59) Height Ratio:	3.13%
Bottom Viewpoint Height (45):	387	Bottom Display Height Ratio (44):	30.23%

Top Composite Video Display (51)						
Viewport Image	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
	x (pixels)	y (pixels)				
Top Left (52)	Top_Left_x	Top_Left_y	Top_Left_Width	Top_Left_Height	Top_Left_FOV	Top_Left_Heading_Ang
	0	0	640	853	70	26.4
Top Center (53)	Top_Center_x	Top_Center_y	Top_Center_Width	Top_Center_Height	Top_Center_FOV	Top_Center_Heading_Ang
	640	0	640	853	70	90.0
Top Right (54)	Top_Right_x	Top_Right_y	Top_Right_Width	Top_Right_Height	Top_Right_FOV	Top_Right_Heading_Ang
	1280	0	640	853	70	153.7

Bottom Composite Video Display (55)						
Viewport Image	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
	x (pixels)	y (pixels)	Bottom_Left_Width	Bottom_Left_Height	Bottom_Left_FOV	Bottom_Left_Heading_Ang
Bottom Left (56)	Bottom_Left_x	Bottom_Left_y				
Bottom Center (57)	Bottom_Center_x	Bottom_Center_y	Bottom_Center_Width	Bottom_Center_Height	Bottom_Center_FOV	Bottom_Center_Heading_Ang
		893				
Bottom Right (58)	Bottom_Right_x	Bottom_Right_y	Bottom_Right_Width	Bottom_Right_Height	Bottom_Right_FOV	Bottom_Right_Heading_Ang
	1280	893				

FIG. 9

TABLE 5

Predetermined Video Display Layout (43) - BVT Layout Type 3

Top Viewpoint Height (45):	960	Top Display Height Ratio (44):	75.00%
Navigation Menu (59) Height:	40	Navigation Menu (59) Height Ratio:	3.13%
Bottom Viewpoint Height (45):	280	Bottom Display Height Ratio (44):	21.88%

Viewport Image	Top Composite Video Display (51)					
	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
Top Left (52)	x (pixels)	y (pixels)				
	Top_Left_x	Top_Left_y	Top_Left_Width	Top_Left_Height	Top_Left_FOV	Top_Left_Heading_Ang
Top Center (53)	0	0	640	960	70	32.0
	Top_Center_x	Top_Center_y	Top_Center_Width	Top_Center_Height	Top_Center_FOV	Top_Center_Heading_Ang
Top Right (54)	640	0	640	960	70	90.0
	Top_Right_x	Top_Right_y	Top_Right_Width	Top_Right_Height	Top_Right_FOV	Top_Right_Heading_Ang
	1280	0	640	960	70	148.1

Viewport Image	Bottom Composite Video Display (55)					
	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
Bottom Left (56)	x (pixels)	y (pixels)				
	Bottom_Left_x	Bottom_Left_y	Bottom_Left_Width	Bottom_Left_Height	Bottom_Left_FOV	Bottom_Left_Heading_Ang
Bottom Center (57)	0	1000	640	280	70	152.3
	Bottom_Center_x	Bottom_Center_y	Bottom_Center_Width	Bottom_Center_Height	Bottom_Center_FOV	Bottom_Center_Heading_Ang
Bottom Right (58)	640	1000	640	280	70	270.0
	Bottom_Right_x	Bottom_Right_y	Bottom_Right_Width	Bottom_Right_Height	Bottom_Right_FOV	Bottom_Right_Heading_Ang
	1280	1000	640	280	70	387.8

FIG. 10

TABLE 6

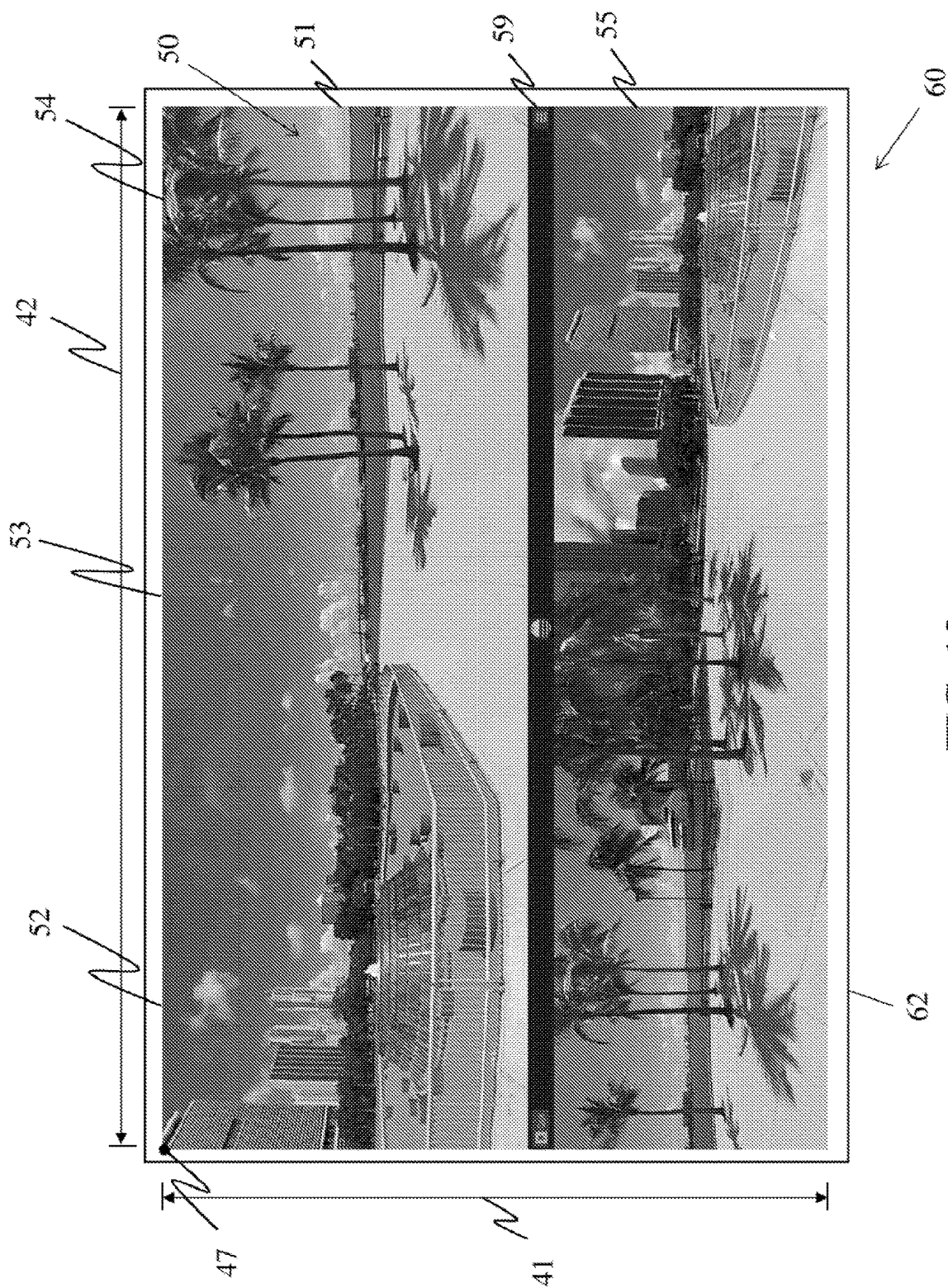
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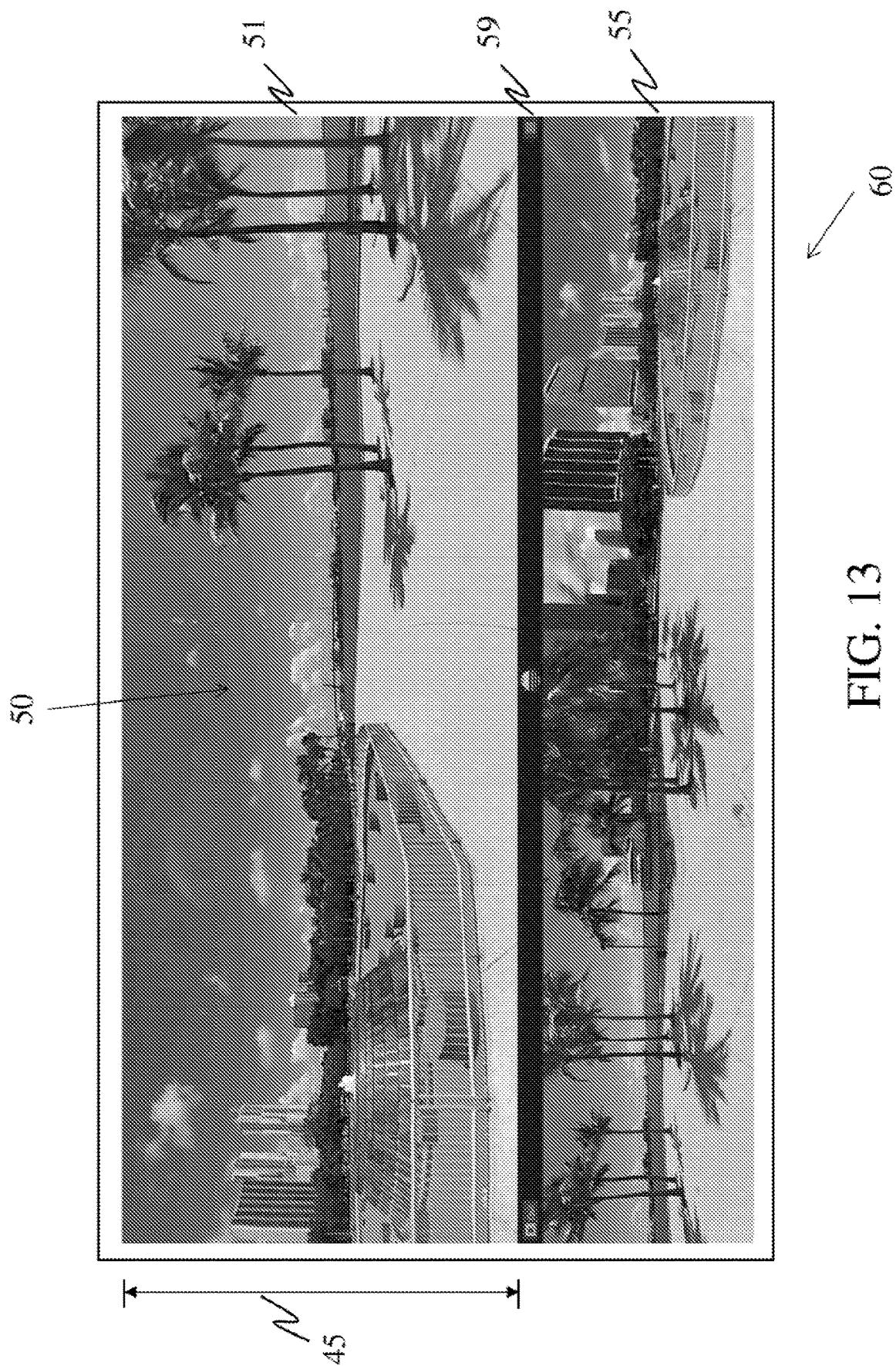
Top Viewpoint Height (45):	1024	Top Display Height Ratio (44):	80.00%
Navigation Menu (59) Height:	40	Navigation Menu (59) Height Ratio:	3.13%
Bottom Viewpoint Height (45):	216	Bottom Display Height Ratio (44):	16.88%

Viewport Image	Top Composite Video Display (51)					
	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
Top Left (52)	x (pixels)	y (pixels)				
	Top_Left_x	Top_Left_y	Top_Left_Width	Top_Left_Height	Top_Left_FOV	Top_Left_Heading_Ang
	0	0	640	1024	70	35.2
Top Center (53)	Top_Center_x	Top_Center_y	Top_Center_Width	Top_Center_Height	Top_Center_FOV	Top_Center_Heading_Ang
	640	0	640	1024	70	90.0
	Top_Right_x	Top_Right_y	Top_Right_Width	Top_Right_Height	Top_Right_FOV	Top_Right_Heading_Ang
	1280	0	640	1024	70	144.9

Viewport Image	Bottom Composite Video Display (55)					
	Viewport Positioning Coordinates (47')		Viewport Width (46) (pixels)	Viewport Height (45) (pixels)	Viewport FOV Angle (48) (degrees)	Viewpoint Heading Angle (49) (degrees)
Bottom Left (56)	x (pixels)	y (pixels)				
	Bottom_Left_x	Bottom_Left_y	Bottom_Left_Width	Bottom_Left_Height	Bottom_Left_FOV	Bottom_Left_Heading_Ang
	0	1064	640	216	90	127.3
Bottom Center (57)	Bottom_Center_x	Bottom_Center_y	Bottom_Center_Width	Bottom_Center_Height	Bottom_Center_FOV	Bottom_Center_Heading_Ang
	640	1064	640	216	90	270.0
	Bottom_Right_x	Bottom_Right_y	Bottom_Right_Width	Bottom_Right_Height	Bottom_Right_FOV	Bottom_Right_Heading_Ang
	1280	1064	640	216	90	412.8

FIG. 11





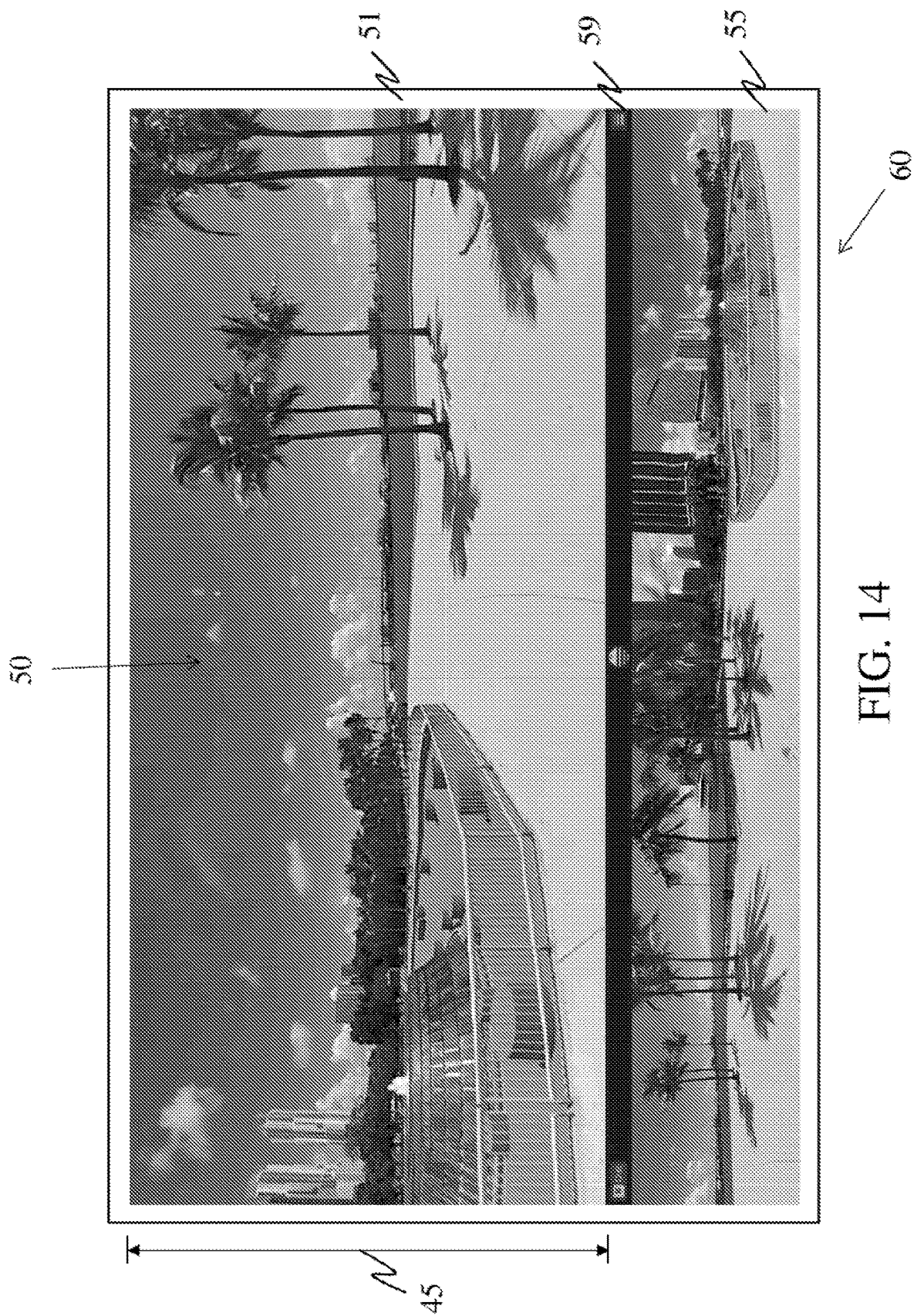
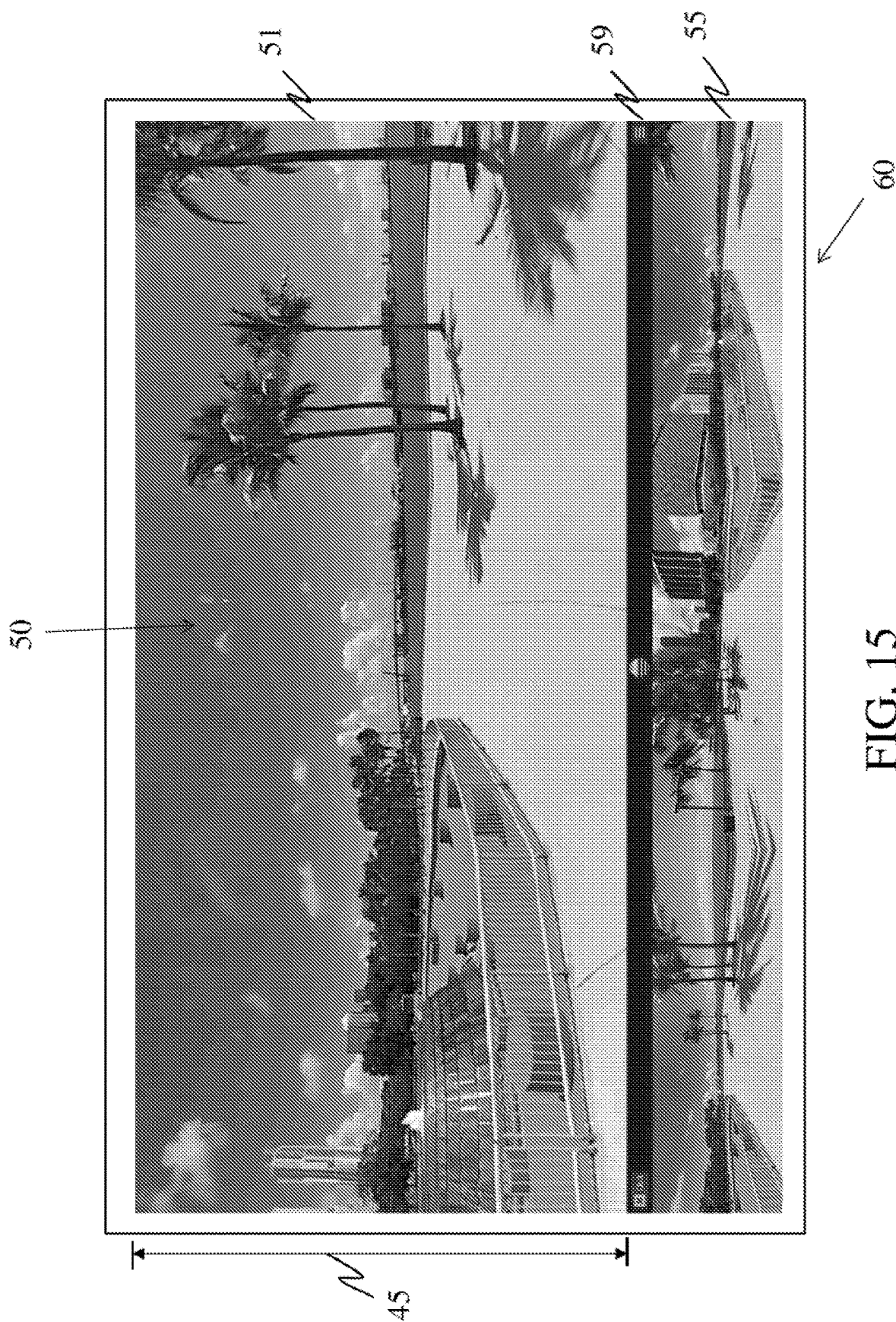


FIG. 14



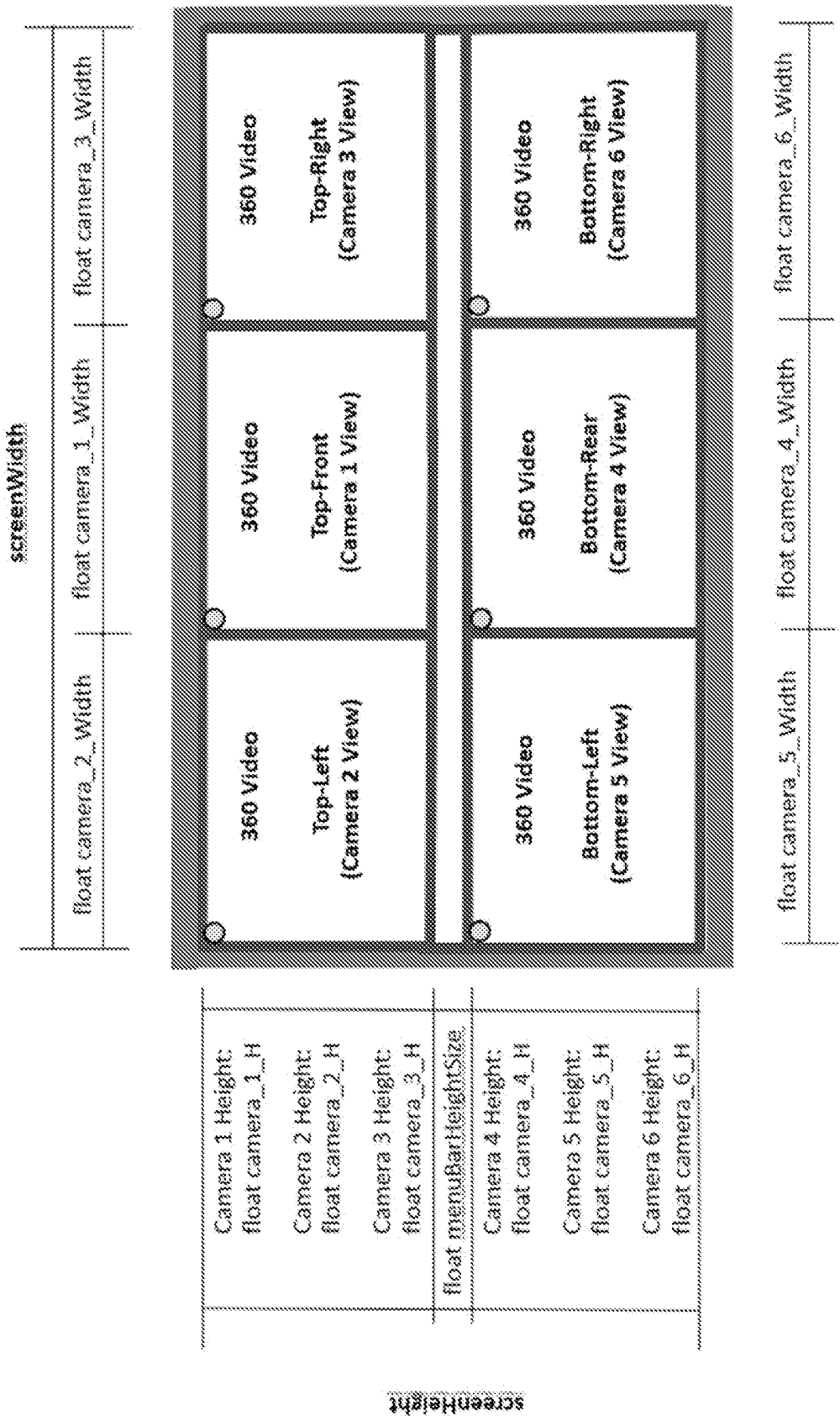


FIG. 16

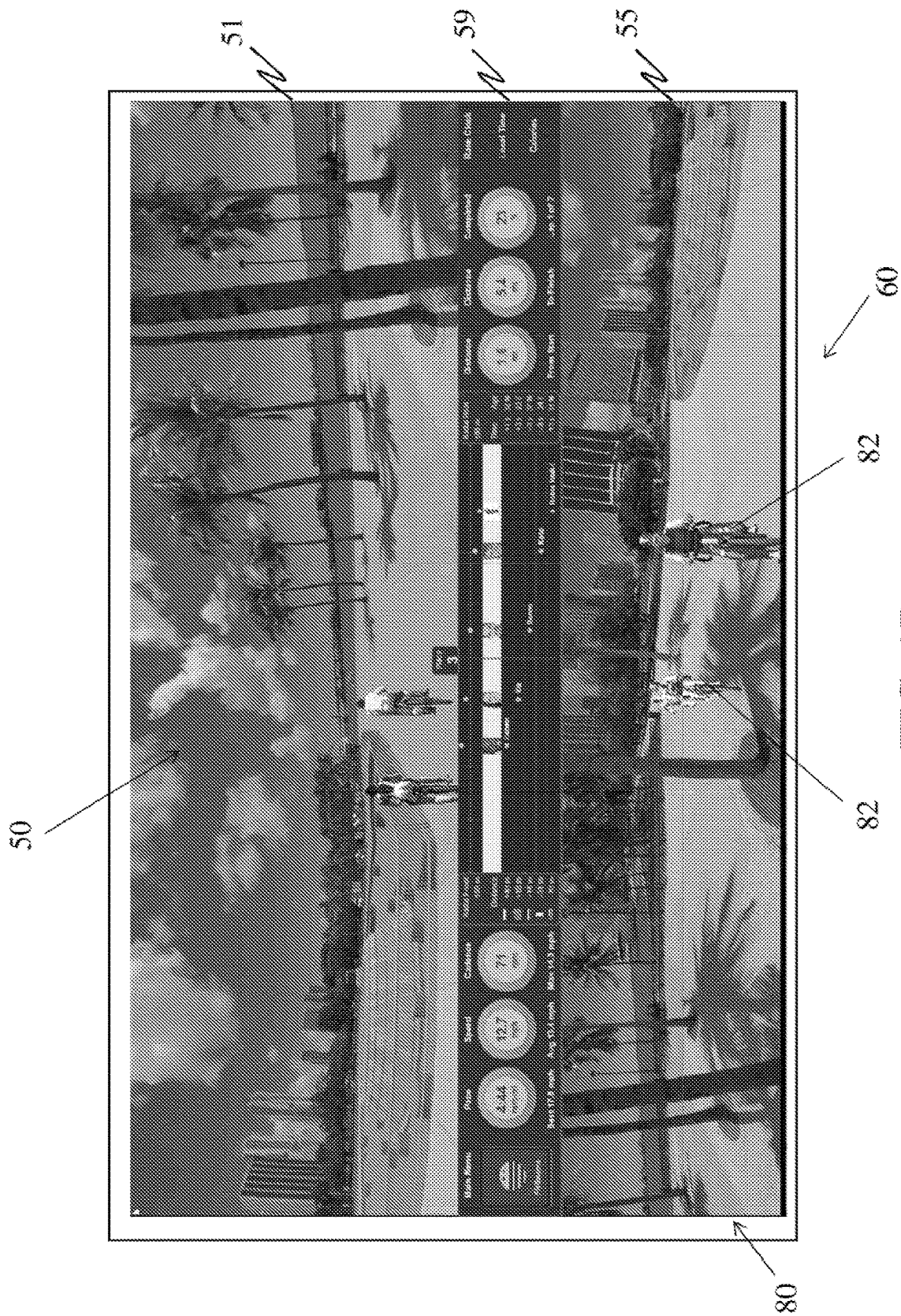
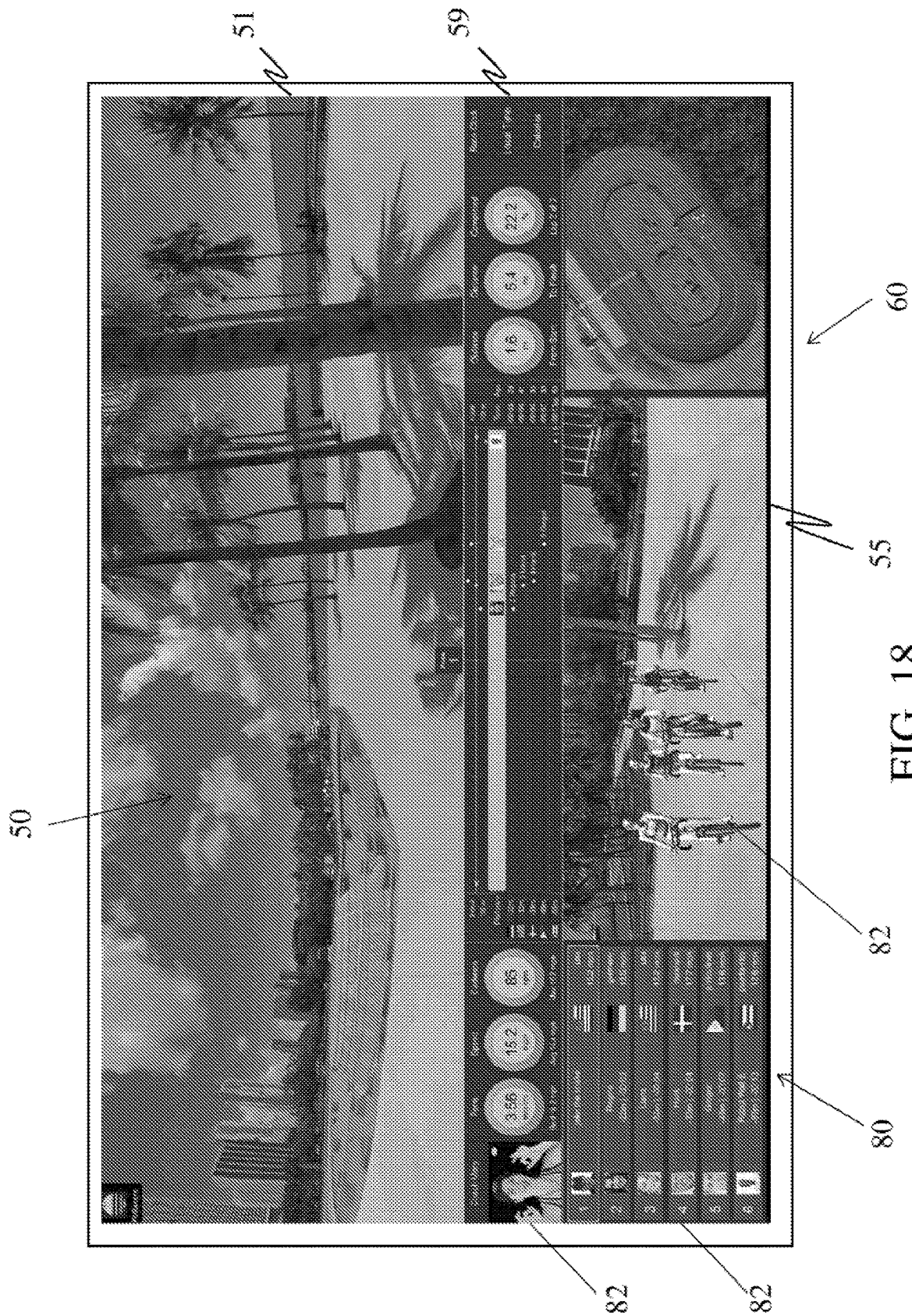
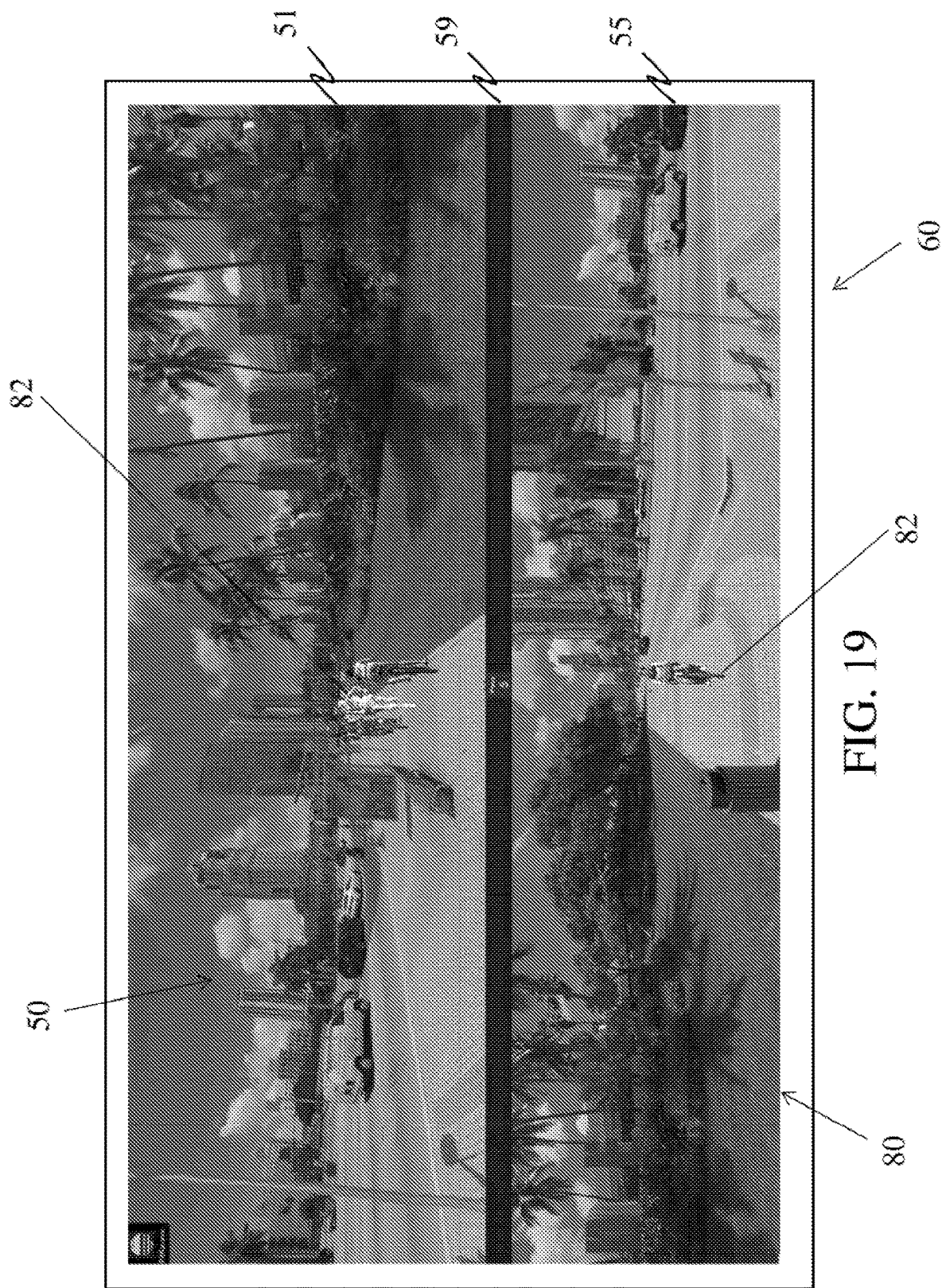
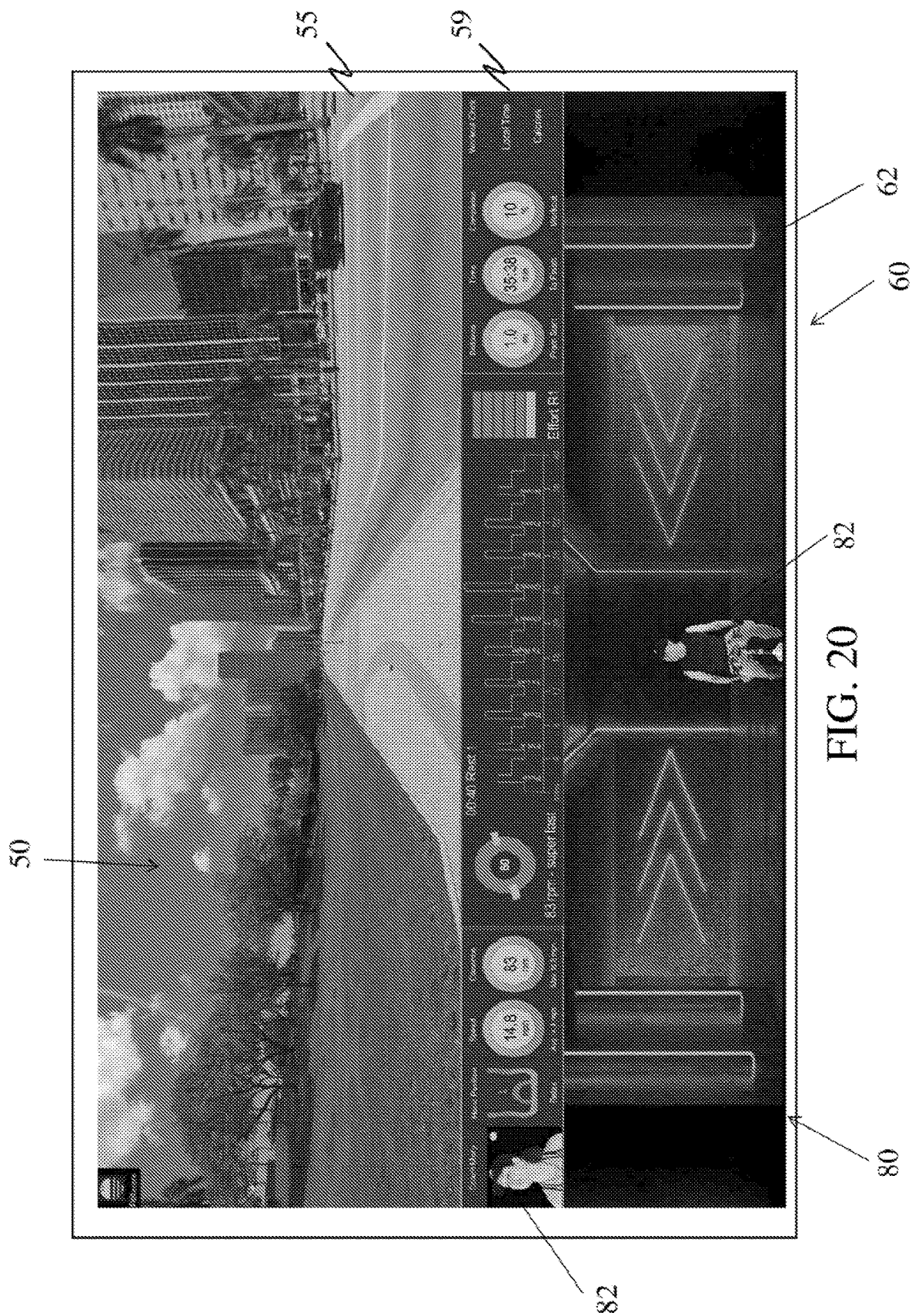


FIG. 17







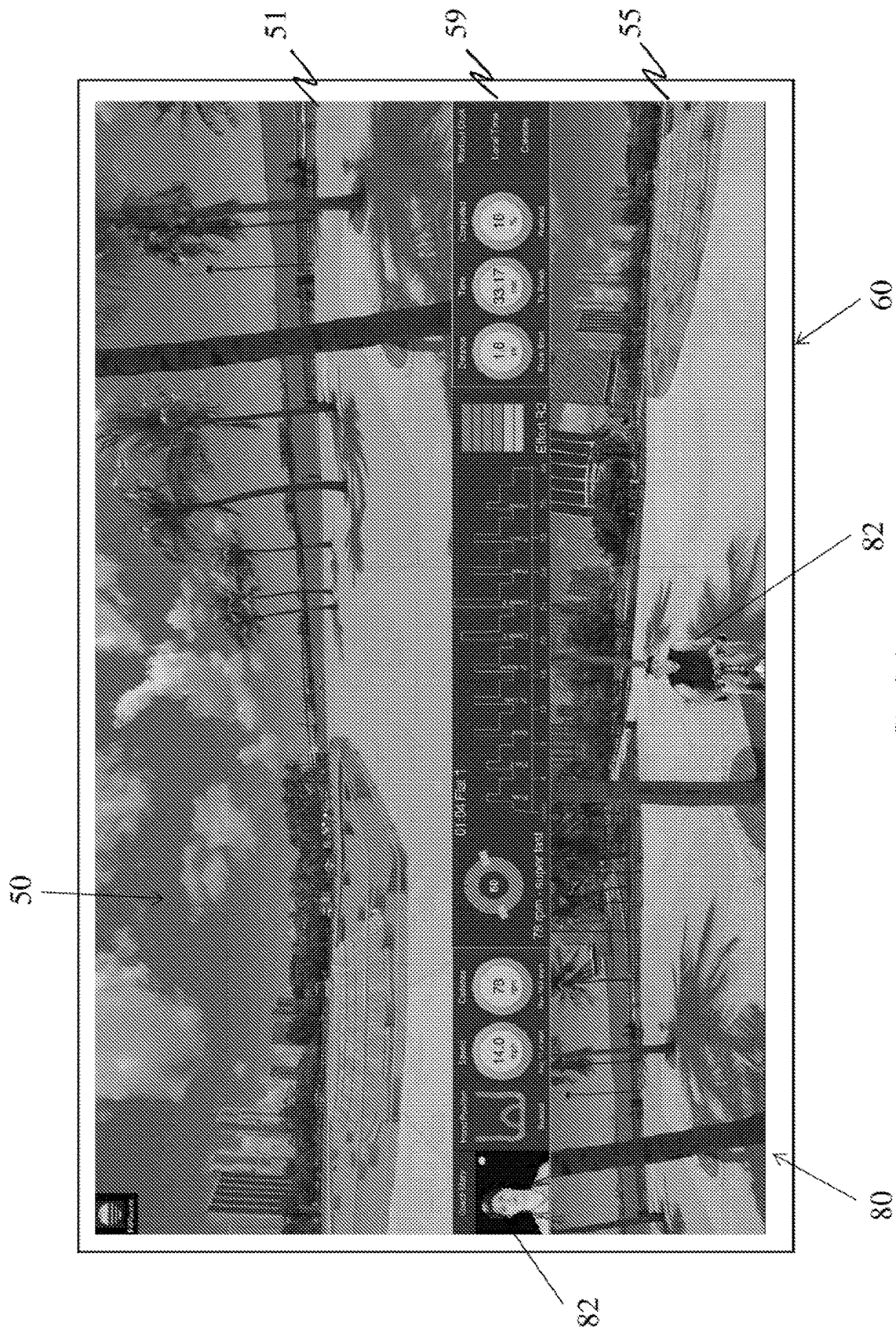


FIG. 21

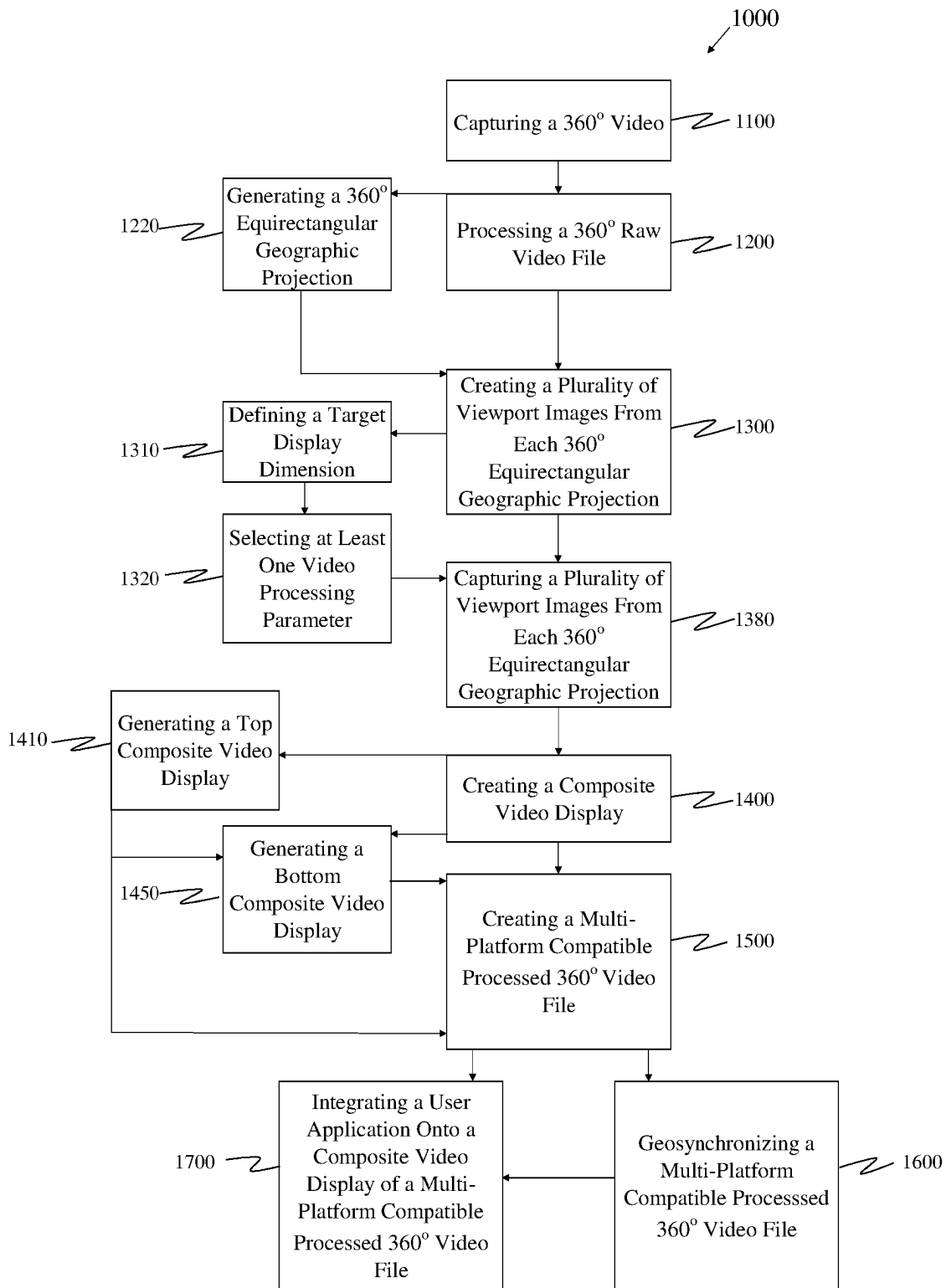


FIG. 22

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SYSTEMS AND METHOD FOR CAPTURING, PROCESSING, AND DISPLAYING A 360° VIDEO

The current application is a continuation application of
the U.S. non-provisional application Ser. No. 16/554,827
filed on Aug. 29, 2019.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention is directed to a system and method
for capturing, processing and displaying a 360° video, and in
particular, a system and method to process a raw 360° video
file such that a full 360° field of view may be viewed by a
user in its entirety on a single video display screen.

Description of the Related Art

There are a number of affordable 360° video cameras on
the market today, making a 360° video somewhat ubiquitous
relative to just a few years ago today. While the technology
required to capture 360° videos has progressed rapidly to
allow this phenomenon to occur, the technology for play-
back and viewing such the plethora of 360° videos has not
yet kept pace. Many 360° video viewers still require a user
to use arrow keys or buttons, or to click and drag his or her
way around a 360° video to view the various limited field of
view perspectives. As is known to anyone who has
attempted such maneuvering, it is easy to lose one's bearings
and sense of the original direction of motion. Further,
whether click and drag or using arrow keys on a computer
keyboard or buttons on a mouse, this process is slow and
tedious.

Further, because currently known viewers utilize a Mer-
cator type projection, essentially stretching a 360° video
around a virtual sphere, resolution is severely degraded,
especially in the polar regions of the virtual sphere.

Virtual reality ("VR") headsets alleviate some of the
aforementioned issues related to viewing a 360° video, such
as navigation, however, VR technology brings with it its
own set of drawbacks. As an initial matter, because the video
screen in a VR headset is positioned mere inches away from
the user's eyes causes strain, and if combined with motion
of the user, for example, while riding a stationary bicycle,
many users quickly become disoriented and/or nauseous.
Further, the weight of the VR headset itself makes it
uncomfortable for users for more than a few minutes,
especially if the user is moving. It is also common for users
to sweat while wearing a VR headset from more than a few
minutes, in some cases, not even that, and this can result in
fogging of the lenses requiring a disruption in whatever
activity the user was engaged.

As such, it would be highly beneficial to provide a 360°
video capture, processing and display system which allows
a user to view an entire 360° panoramic view on a video
display at one time. It would be further advantageous to
generate a processed 360° video file which a user may view
on any of a multiplicity of display platforms, such as, but not
limited to a television screen, a computer monitor, a tablet
computer, a smartphone, etc. Another advantage may be
realized by generating a processed 360° video file which a
user may view without requiring any specialized video
graphics hardware or software, such as a high capacity
computer processor, a high capacity video graphics card, or
a VR headset. Yet one further benefit may be obtained by

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integrating a user application onto such a multi-platform
compatible processed 360° video file.

SUMMARY OF THE INVENTION

The present invention is directed to a 360° video capture,
processing and display system. In at least one embodiment,
the system includes a 360° video camera to capture a 360°
raw video file, and in one further embodiment, the system
comprises a 360° video processing assembly to generate a
plurality of 360° equirectangular geographic projections
from each frame of the 360° raw video file.

In accordance with at least one embodiment of the present
invention, a 360° video processing assembly includes a 360°
video processing algorithm utilizing a plurality of video
processing parameters to capture a plurality of viewport
images from each of the plurality of 360° equirectangular
geographic projections. The present system also includes a
plurality of composite video displays, each composite video
display having a field of view of at least 360°. In accordance
with one embodiment of the present invention, each com-
posite video display comprises a plurality of viewport
images from a corresponding one of each of the plurality of
360° equirectangular geographic projection. Further, each
composite video display in accordance with at least one
embodiment of the present invention includes at least a top
composite video display and a bottom composite video
display, wherein at least one of the top composite video
display or the bottom composite video display comprises a
seamless integration of corresponding ones of the plurality
of viewport images. A multi-platform compatible processed
360° video file comprising the plurality of composite video
displays in generated, and the multi-platform compatible
processed 360° video file is viewable in its entirety on a
video display assembly.

The present invention is further directed to a 360° video
processing system to create a multi-platform compatible
processed 360° video file from a 360° raw video file which
is viewable on a video display screen, wherein the system
includes a 360° video processing assembly which generates
a 360° equirectangular geographic projection from each
frame of the 360° raw video file. In at least one embodiment,
the 360° video processing assembly implements a 360°
video processing algorithm utilizing a plurality of video
processing parameters to capture a plurality of viewport
images from each of a plurality of 360° equirectangular
geographic projections.

In at least one embodiment, the present system generates
a plurality of composite video displays, wherein each of the
composite video displays have a field of view of at least
360°. In one further embodiment, each composite video
display comprises a plurality of viewport images from a
corresponding one of each of said plurality of 360° equirec-
tangular geographic projections.

More in particular, in at least one embodiment, each
composite video display includes at least one of a top
composite video display comprising a top left viewport
image, a top center viewport image and a top right viewport
image or a bottom composite video display comprising a
bottom left viewport image, a bottom center viewport image
and a bottom right viewport image. Further, at least one of
the top composite video display or the bottom composite
video display comprises a seamless integration of corre-
sponding ones of the plurality of viewport images.

As before, in at least one embodiment, the present 360°
video processing system generates a plurality of composite
video displays, wherein each of the composite video dis-

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plays have a field of view of at least 360°. In one further embodiment, each composite video display comprises a plurality of viewport images from a corresponding one of each of said plurality of 360° equirectangular geographic projections.

At least one embodiment of the present invention is directed to a method for capturing, processing and displaying a 360° video on a video display screen, wherein the method includes: capturing a 360° video; generating a plurality of 360° equirectangular geographic projections; selecting one or more of a plurality of video processing parameters, wherein the video processing parameters include a predetermined video display layout, a display height ratio, a viewport field of view angle, and a viewport heading angle; capturing a plurality of viewport images from each of the plurality of 360° equirectangular geographic projections utilizing the one or more video processing parameters; generating a top composite video display from corresponding ones of the plurality of viewport images from each of the plurality of 360° equirectangular geographic projection; generating a bottom composite video display from corresponding ones of the plurality of viewport images from each of the plurality of 360° equirectangular geographic projection; combining the top composite video display and the bottom composite video display from each of the plurality of 360° equirectangular geographic projection to form a composite video display having a field of view of at least 360°; and creating a multi-platform compatible processed 360° video file comprising a plurality of composite video displays, wherein the multi-platform compatible processed 360° video file is viewable on the video display screen.

These and other objects, features and advantages of the present invention will become clearer when the drawings as well as the detailed description are taken into consideration.

BRIEF DESCRIPTION OF THE DRAWINGS

For a fuller understanding of the nature of the present invention, reference should be had to the following detailed description taken in connection with the accompanying drawings in which:

FIG. 1 is a schematic representation of one illustrative embodiment of a 360° video capture, processing and display system in accordance with the present invention.

FIG. 2 is a schematic representation of one illustrative embodiment of a 360° video capture assembly in accordance with the present invention.

FIG. 3 is a schematic representation of one illustrative embodiment of a 360° video processing assembly in accordance with the present invention.

FIG. 4 is a screen shot of one frame of a raw 360° video file in equirectangular video format in accordance with the present invention.

FIG. 5 is a schematic representation of one embodiment of an equirectangular geographic projection in accordance with the present invention.

FIG. 6 is a schematic representation of one embodiment of capturing a plurality of viewport images from an equirectangular geographic projection in accordance with the present invention.

FIG. 7 is a schematic representation of one illustrative embodiment of a plurality of viewport images comprising a composite video display in accordance with the present invention.

FIG. 8 is a table comprising an illustrative embodiment of a plurality of video processing parameters corresponding to

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a predetermined video display layout designated as BVT Layout Type 1 in accordance with the present invention.

FIG. 9 is a table comprising an illustrative embodiment of a plurality of video processing parameters corresponding to a predetermined video display layout designated as BVT Layout Type 2 in accordance with the present invention.

FIG. 10 is a table comprising an illustrative embodiment of a plurality of video processing parameters corresponding to a predetermined video display layout designated as BVT Layout Type 3 in accordance with the present invention.

FIG. 11 is a table comprising an illustrative embodiment of a plurality of video processing parameters corresponding to a predetermined video display layout designated as BVT Layout Type 4 in accordance with the present invention.

FIG. 12 is a screen shot of one illustrative embodiment of a composite video display corresponding to a predetermined video display layout designated as BVT Layout Type 1 in accordance with the present invention.

FIG. 13 is a screen shot of one illustrative embodiment of a composite video display corresponding to a predetermined video display layout designated as BVT Layout Type 2 in accordance with the present invention.

FIG. 14 is a screen shot of one illustrative embodiment of a composite video display corresponding to a predetermined video display layout designated as BVT Layout Type 3 in accordance with the present invention.

FIG. 15 is a screen shot of one illustrative embodiment of a composite video display corresponding to a predetermined video display layout designated as BVT Layout Type 4 in accordance with the present invention.

FIG. 16 is a schematic representation of a composite video display identifying at least some of the variables utilized in the illustrative embodiment of the portion of a 360° video processing algorithm in accordance with the present invention.

FIG. 17 is a screen shot illustrative of one embodiment of an application module incorporating a multi-platform compatible processed 360° video file in accordance with the present invention.

FIG. 18 is a screen shot illustrative of one other embodiment of an application module incorporating a multi-platform compatible processed 360° video file in accordance with the present invention.

FIG. 19 is a screen shot illustrative of one further embodiment of an application module incorporating a multi-platform compatible processed 360° video file in accordance with the present invention.

FIG. 20 is a screen shot illustrative of yet another embodiment of an application module incorporating a multi-platform compatible processed 360° video file in accordance with the present invention.

FIG. 21 is a screen shot illustrative of still one further embodiment of an application module incorporating a multi-platform compatible processed 360° video file in accordance with the present invention.

FIG. 22 is a diagrammatic representation of one illustrative embodiment of a method of capturing, processing and displaying a 360° video in accordance with the present invention.

Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION OF THE INVENTION

In accordance with one embodiment, the present invention is directed to a 360° video capture, processing and

display system, generally as shown as **10** throughout the figures. With reference to the illustrative embodiment of FIG. **1**, the present system **10** includes a 360° video capture assembly **20**. With additional reference to FIG. **2**, a 360° video capture assembly **20** in accordance with the present invention comprises a 360° video camera **22**. A wide variety of 360° video cameras **22** are available today ranging from inexpensive hobbyist 360° video cameras **22** to high end professional 360° video cameras **22**. It is within the scope and intent of the present invention to utilize any type of 360° video camera **22** capable of generating a raw 360° video file **26** of a target scene.

A 360° video capture assembly **20** in accordance with at least one embodiment of the present system **10** further comprises a 360° video camera transport **24**. As shown in the illustrative embodiment of FIG. **2**, a 360° video capture assembly **20** in accordance with the present invention may further comprise a video camera support **23** in order to secure a 360° video camera **22** to a 360° video camera transport **24**.

As will be appreciated by those of skill in the art, a 360° video camera transport **24** in accordance with one embodiment of the present system **10** may comprise any of a plurality of transport devices including but in no manner limited to a manned vehicle such as a bicycle, tricycle, scooter, skateboard, motorcycle, automobile, airplane, helicopter, boat, kayak, submarine, surfboard, paddleboard, rocket, etc., in order to transport a 360° video camera **22** while generating a raw 360° video file of a target scene. In accordance with at least one other embodiment of the present system **10**, a 360° video camera transport **24** may comprise, but is in no manner limited to, an unmanned vehicle such as a remote controlled bike, car, plane, drone, boat, submarine, rocket, etc., once again, in order to transport a 360° video camera **22** during generation of a raw 360° video file of a target scene.

Alternatively, a 360° video camera **22** of a 360° video transport capture assembly **20** in accordance with at least one embodiment of the present invention may be simply handheld by a user as he or she walks or is otherwise transported through a target scene. In one further embodiment, a body camera type mount may be used to attach a 360° video camera **22** to a person as he or she is transported through a target scene. In one further embodiment, a body camera type mount may be used to attach a 360° video camera **22** to an animal, such that the 360° video camera **22** generates a raw 360° video file along and through the course

of the animal's travels. As will be appreciated, animals ranging from cats, dogs, elephants, lions, birds, dolphins, manatees, whales, sharks, etc., just to name a few, may be utilized to "transport" a 360° video camera **22** while generating a raw 360° video file along and through the course of its travels.

As will be further appreciated by those of skill in the art, currently available 360° video cameras **22** are generally capable of capturing a raw 360° video file **26** in 4K resolution, and more specifically, a resolution of either 4096×2160 pixels or 3840×2160 pixels. It is anticipated that 360° video cameras **22** in the immediate future will be capable of capturing raw 360° video files **26** in 8K resolution, 16K resolution, 32K resolution, etc., and it remains within the scope and intent of the present invention to utilize raw 360° video files **26** having such higher resolutions and beyond.

In accordance with at least one embodiment, a 360° video camera **22** will generate a raw 360° video file **26** in an equirectangular 360° video format **37**, such as is shown in the illustrative embodiment of FIG. **4**. Alternatively, a 360° video processing algorithm **36** is employed, at least partially, to convert a raw 360° video file **26**, such as, a fisheye 360° video format or dual fisheye 360° video format, into an equirectangular 360° video format **37**.

Looking once again to the illustrative embodiment of a 360° video capture, processing and display system **10** is shown in FIG. **1**, the system **10** further comprises a 360° video processing assembly **30**. FIG. **3** is a schematic representation of one illustrative embodiment of a 360° video processing assembly **30** in accordance with the present invention. As may be seen from the illustrative embodiment of FIG. **3**, the primary input to the 360° video processing assembly **30** is a raw 360° video file **26**. A plurality of video processing parameters **40**, discussed in greater detail hereinafter, are also input to a 360° video processing algorithm **36** of a 360° video processing assembly **30** in accordance with at least one embodiment of the present invention.

The illustrative embodiment of a 360° video processing assembly **30** as shown in FIG. **3** includes a high-capacity computer processor **32** which is disposed in a communicative relation with a high-capacity video graphics card **34**.

As will be appreciated by those of skill in the art, the specifications presented in Table **1** below are exemplary of a high-capacity computer processor **32** which may be utilized in a 360° video processing assembly **30** in accordance with the present invention.

TABLE 1

Description	Specification
GPU:	GeForce R TX 2080 OC 8 GB
CPU:	Core i7-9700K (4.6GHz Overclocked)
Motherboard:	MicroATX Z370
RAM:	HyperX 32 GB DDR4 2933MHz (2 X 16 GB)
Storage:	512 GB PCIe NVMe M.2 SSD, 2 TB 7200 RPM SATA HDD
PSU:	Alienware 850 Watt Multi-GPU PSU
CPU Cooling:	Alienware High Performance Liquid Cooling System
Core i9-9900K	Core i7-9700K
Architecture:	Coffee Lake
Socket:	1151
Cores / Threads:	8/16
Base Frequency (GHz):	3.6
Boost Frequency	1-2 Cores - 5.0
(ActiveCores - GHz):	
L3 Cache:	4 Cores - 4.8
	8 Cores - 4.7
	16MB
Process:	14 nm++

TABLE 1-continued

Description	Specification
TDP:	95W
Memory Speed:	DDR4-2666
Memory Controller:	Dual-Channel
PCIe Lanes:	x16
Integrated UHD	350/1200
Graphics GT2	
(Base/ Boost MHz):	

As will also be appreciated by those of skill in the art, the specifications presented in Table 2 below are exemplary of a high-capacity video graphics card 34 which may be utilized in a 360° video processing assembly 30 in accordance with the present invention.

TABLE 2

GEFORCE RTX 2080 FOUNDERS EDITION - Reference Product Specs:	
GPU Engine Specs:	
29442944 NVIDIA CUD A ® Cores	
57160 TRTX-OPS	
88 Giga Rays/s	
17101800 (QC) Boost Clock (MHz)	
15151515 Base Clock (MHz)	
Memory Specs:	
14 Gbps Memory Speed	
8 GB GDDR6 8 GB GDDR6 Standard Memory Configuration	
256-bit Memory Interface Width	
448 GB/s Memory Bandwidth (GB/sec)	

With continued reference to the 360° video processing assembly 30 as shown in FIG. 3, the high-capacity computer processor 32 and the high-capacity video graphics card 34 are disposed in an operative communication with a 360° video processing algorithm 36. Following this Detailed Description in the present specification is an exemplary portion of a 360° video processing algorithm 36 in accordance with at least one embodiment of the present invention.

In at least one embodiment, the 360° video processing algorithm 36 operates to convert a raw 360° video file 26 into an equirectangular 360° video format 37, once again, as shown in the illustrative embodiment of FIG. 4. In at least one further embodiment, the 360° video processing algorithm 36 operates to generate an equirectangular 360° geographic projection 37', such as is shown by way of example in the illustrative embodiment of FIG. 5, from each frame of the raw 360° video file 26 in an equirectangular 360° video format 37.

As stated above, the present 360° video capture, processing and display system 10 further comprises a plurality of video processing parameters 40. More in particular, in at least one embodiment, the plurality of video processing parameters 40 are utilized by the 360° video processing algorithm 36 to precisely define each of a plurality of viewport images 52, 53, 54 and 56, 57, 58 to be captured from an equirectangular 360° geographic projection 37' for each frame of a raw 360° video file 26. FIGS. 6 and 7 are illustrative of a plurality of viewport images 52, 53, 54 and 56, 57, 58 captured from a single equirectangular 360° geographic projection 37' of a raw 360° video file 26 in an equirectangular 360° video format 37.

In at least one embodiment, the video processing assembly 30 captures each of a plurality of viewport images 52,

53, 54 and 56, 57, 58 from each 360° equirectangular geographic projection 37' at a rate of about thirty frames per second. In at least one further embodiment, the video processing assembly 30 captures each of a plurality of viewport images 52, 53, 54 and 56, 57, 58 from each 360° equirectangular geographic projection 37' at a rate of about sixty frames per second. As will be appreciated by those of skill in the art, the capture rate, whether thirty frames per second or sixty frames a second, is a function of the current state of the art in video processing, however, it is understood to be within the scope and intent of the present invention to employ capture rates that are less than thirty frames per second or that are greater than sixty frame per second, and which may be substantially greater than sixty frame per second.

The plurality of video processing parameters 40 in accordance with at least one embodiment of a 360° video processing assembly 30 in accordance with the present invention includes a target display width 42 and a target display height 41. In at least one embodiment, a target display width 42 and a target display height 41 are expressed in terms of a number of pixels.

In accordance with at least one further embodiment of a 360° video processing assembly 30 in accordance with the present invention, the plurality of video processing parameters 40 include a plurality of predetermined video display layouts 43. Each predetermined video display layout 43 is at least partially defined by a display height ratio 44. More in particular, a display height ratio 44 is at least partially defined as a height 45 of a viewport image 52, 53, 54 and 56, 57, 58 divided by the target display height 41, with the quotient multiplied by 100. As one example, for a predetermined video display layout 43 designated as BVT Layout Type 1, a display height ratio 44 is calculated to be 58.98%. This display height ratio 44 is obtained by dividing a top viewport height 45 of 755 pixels by a target display height of 1280 pixels, and multiplying the quotient by 100.

In addition to a display height ratio 44 and a viewport height 45, the plurality of video processing parameters 40 in accordance with at least one embodiment of a 360° video processing assembly 30 include a viewport width 46, viewport origin coordinates 47, viewport positioning coordinates 47', a viewport field-of-view angle 48, and/or a viewport center heading angle 49.

Tables 3 through 6 presented in FIGS. 8 through 11, respectively, assign specific video processing parameters 40 for each of a plurality of predetermined video display layouts 43 designated as BVT Layout Types 1 through 4, respectively, in accordance with at least one embodiment of the present 360° video capture, processing and display system 10 of the present invention. Each of the examples in Tables 3 through 6 are based on a target display width 42 of 1920 pixels and a target display height 41 of 1280 pixels.

Turning next to the illustrative embodiment of FIG. 8, Table 3 assigns the specific video processing parameters 40

for a predetermined video display layout 43 designated as BVT Layout Type 1. Once again, the plurality of video processing parameters 40 presented in Tables 3 through 6 are based on a target display width of 1920 pixels and a target display height 41 of 1280 pixels. As such, Table 3 assigns a top viewport height 45 of 755 pixels, which corresponds to a top display height ratio 44 of 58.98%, calculated as shown above. Table 3 further assigns a bottom viewport height 45 of 485 pixels, which corresponds to a bottom display height ratio 44 of 37.89%. The remaining 40 pixels of the target display height 41 are utilized to display a navigation menu 59.

Table 3 further assigns a viewport width 46 of 640 pixels for each of viewport images 52, 53, 54 and 56, 57, 58. Likewise, Table 3 assigns a viewport field-of-view angle 48 of 70° for each of viewport images 52, 53, 54 and 56, 57, 58. Looking again to the plurality of video processing parameters 40 in Table 3, viewport positioning coordinates 47 are assigned for each of viewport images 52, 53, 54 and 56, 57, 58. In at least one embodiment of the present invention, viewport positioning coordinates 47' are expressed as x, y coordinates at the uppermost left-hand corner of the corresponding viewport image 52, 53, 54 and 56, 57, 58, and are measured in pixels from viewport origin coordinates 47. More in particular, viewport origin coordinates 47 are expressed as x=0 pixels and y=0 pixels, with the viewport origin coordinates 47 located at the uppermost left-hand corner of a composite video display 50, such as is shown by way of example in the illustrative embodiments of FIGS. 7 and 12.

Looking further to the plurality of video processing parameters 40 in Table 3 in FIG. 8, for a predetermined video display layout 43 designated as BVT Layout Type 1, top left viewport image 52 is assigned viewport positioning coordinates 47' wherein x=0 pixels and y=0 pixels; top center viewport image 53 is assigned viewport positioning coordinates 47' wherein x=640 pixels and y=0 pixels; and, top right viewport image 54 is assigned viewport positioning coordinates 47' wherein x=1280 pixels and y=0 pixels.

Lastly, Table 3 assigns a viewport center heading angle 49 for each of viewport images 52, 53, 54 and 56, 57, 58. It is noteworthy from a review of Tables 3 through 6 presented in FIGS. 8 through 11, respectively, that the viewport center heading angle 49 for each of the top left viewport image 52, top right viewport image 54, bottom left viewport image 56, and a bottom right viewport image 58 are completely unique for each of the plurality of predetermined video display layouts 43. As may also be seen from Tables 3 through 6, the viewport center heading angle 49 for each of the top center viewport image 53 and the bottom center viewport image 57, are 90° and 270°, respectively.

Application of the plurality of video processing parameters 40 in Table 3 of FIG. 8 by a 360° video processing assembly 30 in accordance with the present invention results in the capture of a plurality of top viewport images 52, 53 and 54 and bottom viewport images 56, 57 and 58, such as is shown by way of example in the illustrative embodiment of FIG. 7. More importantly, application of the specific plurality of video processing parameters 40 of Table 3 to an equirectangular 360° geographic projection 37' of the equirectangular 360° video format 37 frame as shown in the illustrative embodiment of FIG. 4 results in the creation of the composite video display 50, as shown by way of example in the illustrative embodiment of FIG. 12.

It is noteworthy, as may be seen from the illustrative embodiment of FIG. 12 that the present 360° video processing assembly 30 creates a top composite video display 51

comprising a seamless integration of corresponding ones of the top left viewport image 52, top center viewport image 53 and top right viewport image 54. Similarly, as may also be seen from the illustrative embodiment of FIG. 12, the present 360° video processing assembly 30 creates a bottom composite video display 55 comprising a seamless integration of corresponding ones of the bottom left viewport image 56, bottom center viewport image 57 and bottom right viewport image 58.

It is further noteworthy that in accordance with at least one embodiment of the present 360° video capture, processing and display system 10, a composite video display 50 comprising an entire 360° field of view is completely displayed at one time on a video display assembly 60.

In fact, and with further reference to the illustrative embodiment of FIG. 12, the composite video display 50 comprising a composite field of view which is greater than 360° is completely displayed at one time on a flat video display screen 62 of a video display assembly 60 in accordance with the present invention. More in particular, in accordance with the plurality of video processing parameters 40 as shown in Table 3 of FIG. 8, each of the top viewport images 52, 53 and 54 and the bottom viewport images 56, 57, and 58 comprise a viewport field of view angle 48 of 70°. As such, each of top composite video display 51 and bottom composite video display 55 a top or bottom composite field of view of view of 210°, respectively. It of course follows that composite video display 50 comprises a composite field of view of 420°. It is believed that the composite video display 50 having a composite field of view of 360° or greater displayed on a flat video display screen 62 of a video display assembly 60 in accordance with the present invention is heretofore unknown. As will be appreciated, a video display assembly 60 in accordance with at least one embodiment of the present invention may comprise a curved video display screen 62.

As will be appreciated by those of skill in the art, by capturing and compiling a plurality of viewport images 52, 53, 54 and 56, 57, 58 for each of a plurality of successive frames of a raw 360° video file to create a plurality of composite video displays 50, a processed 360° video file may be created. More in particular, and in accordance with at least one embodiment of the present invention, the video processing assembly 30 is utilized to create a multi-platform compatible processed video file 38, as shown in the illustrative embodiment of FIG. 3.

Specifically, a multi-platform compatible processed video file 38 may be in any of a number of common video file formats including, but in no manner limited to, "mp4", "mov", "avi", etc. As such, a multi-platform compatible processed video file 38 created in accordance with the present invention comprises a plurality of composite video displays 50 which are viewable on any of a number of video display assembly 60 platforms including, but in no manner limited to, high definition televisions, standard resolution televisions, computers monitors, laptops, tablets, smartphones screens, etc., just to name a few, without the need for any specialized video graphics hardware or software.

In at least one embodiment, the resultant multi-platform compatible processed video file 38 is output in HD resolution, i.e., about half of the resolution of a raw 360° video file, at a rate of about thirty frames per second to about sixty frames per second. Of course, it will be appreciated by those of skill in the art, as higher resolution raw 360° video files become more readily available, higher resolution multi-platform compatible processed video files 38 may be created utilizing the present video capture, processing and display

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system 10. More in particular, as a result of the data compression that occurs utilizing a 360° video processing assembly 30 in accordance with the present system 10, the resulting multi-platform compatible processed video files 38 may be displayed with greater resolution than that which

may be obtained utilizing currently known 360° video viewing platforms such as, by way of example, Google Streetview, YouTube 360° video, VIRE 360° Video Player, or GoPro Video Player, just to name a few. It is further noteworthy, that as a result of capturing a plurality of viewport images 52, 53, 54, and/or 56, 57, 58, which in total comprise only a portion of the data of a frame of a raw 360° video file 26 image file utilized to generate an equirectangular 360° geographic projection 37'. More in particular, the north and south polar regions of the equirectangular 360° geographic projection 37' are not utilized while capturing a plurality of viewport images 52, 53, 54, and/or 56, 57, 58, as may be seen from the illustrative embodiment of FIG. 6, and therefore, a multi-platform compatible processed video files 38 created in accordance with the present invention comprises a file size that is considerably smaller than that of the raw 360° video file 26 from which it was generated. In accordance with at least one embodiment of the present 360° video capture, processing and display system 10, a multi-platform compatible processed video files 38 comprises a file size which is approximately four times less than that of the raw 360° video file 26 from which it was generated.

Turning next to the illustrative embodiment of FIG. 13, application of the specific plurality of video processing parameters 40 of Table 4 as shown in FIG. 9 to an equirectangular 360° geographic projection 37' of the single equirectangular 360° video format 37 frame as shown in the illustrative embodiment of FIG. 4 results in the composite video display 50 as shown in the illustrative embodiment of FIG. 13. Similarly, application of the specific plurality of video processing parameters 40 of Table 5 as shown in FIG. 10 to an equirectangular 360° geographic projection 37' of the single equirectangular 360° video format 37 frame as shown in the illustrative embodiment of FIG. 4 results in the composite video display 50 as shown in the illustrative embodiment of FIG. 14.

Additionally, application of the specific plurality of video processing parameters 40 of Table 6 as shown in FIG. 11 to an equirectangular 360° geographic projection 37' of the single equirectangular 360° video format 37 frame as shown in the illustrative embodiment of FIG. 4 results in the composite video display 50 as shown in the illustrative embodiment of FIG. 15. As may be seen from Table 6 in FIG. 11, the viewport field of view angle 48 assigned to each of the bottom left viewport image 56, bottom center viewport image 57 and bottom right viewport image 58 is 90°. As such, bottom composite video display 55 in the illustrative embodiment of FIG. 14 comprises a bottom composite field of view of 270°, and further, composite video display 50 in the illustrative embodiment of FIG. 14 comprises a composite field of view of 480°. As will be appreciated by those of skill in the art, in at least one embodiment, the present system 10 may be utilized to generate a composite video display 50 having a field of view of less than 360°.

Turning once again to the 360° video processing assembly 30 as shown in the illustrative embodiment of FIG. 3, a geographical positioning coordinate file 39 may be generated in conjunction with the creation of a multi-platform compatible 360° video file 38. The geographical positioning coordinate file includes geographical positioning coordinates which correspond to each of a plurality of captured

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viewport images, and may include at least a latitude, longitude, and a heading angle in which the lens of the 360° video camera was directed at the time each frame of a raw 360° video was recorded.

In at least one embodiment, a geosynchronization module 70 disposed in communication with a video processing assembly 30 correlates the data in a geographical positioning coordinate file 39 to each of the plurality of captured viewport images corresponding to each frame of a raw 360° video file.

As further shown in the illustrated embodiment of FIG. 3, the present 360° video capture, processing and display system 10 may include a user application module 80. More in particular, an application module 80 is operable to integrate a user application onto a multi-platform compatible 360° video file 38. As discussed hereinafter, there are any number of user applications which may benefit from integration with a multi-platform processed 360° video file 38 in accordance with the present invention including but not limited to racing, touring, exploring, travelogues, movies, real estate, inspections, engineering, surveying, drone videography, security, surveillance, law enforcement, and military applications.

FIGS. 17 through 21 are illustrative of one example of a user application module 80 integrated onto a composite video display 50 of a multi-platform processed 360° video file 38 in accordance with the present invention. More in particular, the illustrative embodiment of a FIGS. 17 through 21 are representative of a user application module 80 comprising a user fitness application. More in particular, user application module 80 comprises a plurality of user application elements 82 which are overlain onto the composite video display 50. As one example, in FIG. 17, a plurality of user application elements 82 in the form of avatars representing participants in a virtual bicycle race relative to a user viewing the video display assembly. As further shown in FIG. 17, the navigation menu 59 comprises the user's personal metrics as well as those comprising the course, and a leaderboard showing the current position for each of the virtual participants relative to the user. An appropriate sensor assembly may be utilized by a user application module 80 so as to measure the actual performance of a user on a stationary cycle, or other such exercise device which may then be translated and displayed on a navigation menu 59, as shown in the illustrative embodiments of FIGS. 17 through 21. An interactive device 64, such as, by way of example only, a remote controller or mouse, may be utilized to facilitate a user's interaction with a user application module 80 via a navigation menu 59.

FIG. 18 is illustrative of a user application module 80 in an alternative composite video display 50, wherein the user application elements 82 further comprise a virtual coach to advise a user on his or her status in the virtual race, as well as strategies to improve his or her performance therein. FIG. 19 is illustrative of a user application module 80 in yet another alternative composite video display 50, wherein the navigation menu 59 has been minimized in the composite video display 50.

The illustrative embodiments of FIGS. 20 and 21 are representative examples of a user application module 80 comprising a solo fitness routine for a user, specifically, a spinning exercise regimen. More in particular, in the illustrative embodiment of FIG. 20, one user application element 82 includes an animated image of the user displayed in the lower portion of the video display screen 62, while a scenic view is presented for the user to view as he or she rides their stationary cycle or similar exercise device. As will be

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appreciated by those of skill in the art, it is within the scope and intent of the present invention to incorporate a live video image of the user as a user application element **82** projected onto a composite video display **50**. As before, the navigation menu **59** includes statistics about the user's performance, and a virtual coach to prompt the user to perform. In the illustrative embodiment of FIG. **21**, the plain background behind the animated image of the user **82** has been replaced with a bottom composite video display **55** to provide the user with the sensation of cycling outdoors.

Turning now to FIG. **22**, one illustrative embodiment of a method for capturing, processing and displaying a 360° video **1000** is presented. As may be seen from FIG. **22**, the present method **1000** begins with capturing a 360° video **1100**. In at least one embodiment, this further entails generating a raw 360° video file.

The present method for capturing, processing and displaying a 360° video **1000** further comprises processing a raw 360° video file **1200**. As noted above, in at least one embodiment, the present method **1000** includes converting a raw 360° video file into a 360° equirectangular video format. Looking again to the illustrative embodiment of FIG. **22**, the present method **1000** also includes generating a 360° equirectangular geographic projection from each frame of a raw 360° video file **1220**.

With continued reference to the illustrative embodiment of FIG. **22**, the present method for capturing, processing and displaying a 360° video **1000** includes creating a plurality of viewport images from each 360° equirectangular geographic projection **1300**. As before, a plurality of video processing parameters are input to a 360° video processing assembly in order to create a plurality of viewport images **1300**. In at least one embodiment, the plurality of video processing parameters include a target display dimension, and in at least one further embodiment, the plurality of video processing parameters include a target display width and a target display height.

At least one further embodiment of the present method for capturing, processing and displaying a 360° video **1000** comprises selecting at least one video processing parameter **1320**. Also as before, the plurality of video processing parameters may include one or more of a predetermined video display layout, a display height ratio, a viewport width, a viewport height, viewport origin coordinates, viewport positioning coordinates, a viewport field of view angle and/or a viewport center heading angle. Of course, it will be appreciated that in at least one embodiment of the present method **1000**, all of the foregoing plurality of video processing parameters are utilized by a 360° video processing assembly to create a plurality of viewport images from each 360° equirectangular geographic projection **1300**, in accordance with the present invention.

Looking once again to the illustrative embodiment of FIG. **22**, the present method for capturing, processing and displaying a 360° video **1000** also includes capturing each of a plurality of viewport images from each 360° equirectangular geographic projection **1380**. As before, in at least one embodiment, the present method **1000** includes capturing each of a plurality of viewport images from each 360° equirectangular geographic projection **1380** at a rate of about thirty frames per second. In at least one further embodiment, the present method **1000** includes capturing each of a plurality of viewport images from each 360° equirectangular geographic projection **1380** at a rate of about sixty frames per second. As will be appreciated by those of skill in the art, the capture rate, whether thirty frames a second or sixty frames a second, is a function of the

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current state of the art in video processing, however, it is understood to be within the scope and intent of the present invention to employ capture rates that are less than thirty frames per second or greater than sixty frames per second, including substantially greater than sixty frames per second.

In at least one further embodiment, the present method **1000** also includes capturing a plurality of geographical positioning coordinates corresponding to each frame of a raw 360° video, which may be correlated to each of the plurality of captured viewport images. In at least one embodiment, the geographical positioning coordinates corresponding to each of the plurality of captured viewport images include at least a latitude, longitude, and a heading angle in which the lens of the 360° video camera was directed at the time each frame of a raw 360° video was recorded.

After capturing a plurality of viewport images from each 360° equirectangular geographic projection **1380**, the present method comprises creating a composite video display **1400** in accordance with at least one embodiment, the present method **1000** includes generating a top composite video display **1410**. In at least one further embodiment, generating a top composite video display **1410** includes positioning a top left viewport image, a top center viewport image, and a top right viewport image in such a manner so as to create a seamless integration of the plurality of top viewport images to generate a top composite video display **1410**. Similarly, in at least one embodiment, the present method **1000** includes generating a bottom composite video display **1450**. As with the top composite video display, generating a bottom composite video display **1450** includes positioning a bottom left viewport image, a bottom center viewport image, and the top right viewport image in such a manner so as to create a seamless integration of the plurality of bottom viewport images to generate a bottom composite video display **1450**. As may be seen from the illustrative embodiment of FIG. **22**, creating a composite video display **1400** may include generating a top composite video display **1410**, or generating a bottom composite video display **1450**, or both.

At least one further embodiment of the present method for capturing, processing and displaying a 360° video **1000** further comprises combining a top composite video display and a bottom composite video display to generate a composite video display, such as those shown by way of example only in the illustrative embodiments of FIGS. **12** through **15**.

With reference once again to the illustrative embodiment of FIG. **22**, the present method **1000** further includes creating a multi-platform compatible processed 360° video file **1500**. As before, a multi-platform compatible processed 360° video file may be an "mp4" file, a "movfile, an "avi" file, etc. As such, and as disclosed and described above, creating a multi-platform compatible processed 360° video file **1500** in accordance with the present method **1000** results in a video file which may be played on any of a plurality of video display assemblies from the screen of a smart phone to an ultrahigh definition video monitor.

With continued reference to the illustrative embodiment of FIG. **22**, the present method for capturing, processing and displaying a 360° video **1000** in at least one embodiment includes geosynchronizing the processed 360° video file **1600**. More in particular, geosynchronizing the processed 360° video file **1600** comprises correlating the plurality of geographical positioning coordinates for each frame of a raw 360° video with a corresponding one of each of the plurality of captured viewport images of the multi-platform compatible processed 360° video file.

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In at least one further embodiment, the present method for capturing, processing and displaying a 360° video **1000** also includes integrating a user application onto a multi-platform processed 360° video file **1700**. As will be appreciated by those of skill in the art, the number of useful applications which may benefit from integration with a multi-platform processed 360° video are essentially limitless. Examples of user applications which may benefit from integration with a multi-platform processed 360° video in accordance with the present invention include racing, touring, exploring, travel-ogues, movies, real estate, inspections, engineering, surveying, dronevideography, security, surveillance, law enforcement and military applications. The present invention lends itself to integration into numerous user application including, by way of example, those wherein navigation with a mouse or other such device is not practical or convenient; applications which are not amenable to integration into virtual reality projections; as well as applications that do not require specialized equipment or training, thus making them more acceptable to a wider segment of society.

In still one further embodiment, the present method for capturing, processing and displaying a 360° video **1000** also includes integrating a user application onto a multi-platform processed 360° video file **1700**, wherein the multi-platform processed 360° video file has been geosynchronized, as disclosed hereinabove.

Since many modifications, variations and changes in detail can be made to the described embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalents.

What is claimed is:

1. A 360° video capture, processing and display system comprising:

- a 360° video camera to capture a 360° raw video file;
- a 360° video processing assembly to generate a plurality of 360° equirectangular geographic projections from said 360° raw video file;
- said 360° video processing assembly comprising a 360° video processing algorithm utilizing a plurality of video processing parameters to capture a plurality of viewport images from each of said plurality of 360° equirectangular geographic projections;
- a plurality of composite video displays;
- each said composite video display comprising said plurality of viewport images from a corresponding one of each of said plurality of 360° equirectangular geographic projection;
- each said composite video display including a top composite video display, said top composite video display comprising a seamless integration of corresponding ones of said plurality of viewport images;
- said top composite video display comprising a plurality of viewport images corresponding to a portion of the 360° equirectangular geographic projections;
- said top composite video display comprising a field of view of 210°;
- a multi-platform compatible processed 360° video file comprising said plurality of composite video displays; and
- said multi-platform compatible processed 360° video file viewable on a video display assembly.

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2. The system as recited in claim 1 wherein said top composite video display comprises a top left viewport image, a top center viewport image and a top right viewport image.

3. The system as recited in claim 1 wherein said plurality of video processing parameters comprises a predetermined video display layout.

4. The system as recited in claim 1 wherein said plurality of video processing parameters comprises a display height ratio.

5. The system as recited in claim 1 wherein said plurality of video processing parameters comprises a viewport field of view angle.

6. The system as recited in claim 1 wherein said plurality of video processing parameters comprises a viewport center heading angle.

7. The system as recited in claim 1 wherein said 360° video processing assembly captures a plurality of viewport images from each of said plurality of 360° equirectangular geographic projections at a rate of about thirty frames per second to about sixty frames per second.

8. The system as recited in claim 1 wherein said 360° video processing assembly captures a plurality of geographical positioning coordinates corresponding to each of said plurality of viewport images captured from each of said plurality of 360° equirectangular geographic projections.

9. The system as recited in claim 8 wherein said plurality of geographical positioning coordinates comprises a latitude, a longitude and a camera heading angle for each of said plurality of viewport images.

10. The system as recited in claim 9 further comprising a geographical positioning coordinate file comprising said plurality of geographical positioning coordinates corresponding to each of said plurality of captured viewport images.

11. A 360° video processing system to create a multi-platform compatible processed 360° video file from a 360° raw video file which is viewable on a video display screen, said system comprising:

- a 360° video processing assembly which generates a 360° equirectangular geographic projection from the 360° raw video file;
- said 360° video processing assembly comprising a 360° video processing algorithm utilizing a plurality of video processing parameters to capture a plurality of viewport images from each of a plurality of 360° equirectangular geographic projections;
- a plurality of composite video displays;
- each said composite video display comprising said plurality of viewport images from a corresponding one of each of said plurality of 360° equirectangular geographic projections;
- each said composite video display including a top composite video display comprising a top left viewport image, a top center viewport image and a top right viewport image, said top composite video display comprising a seamless integration of corresponding ones of said plurality of viewport images;
- the top left viewport image, the top center viewport image and the top right viewport image corresponding to a portion of the 360° equirectangular geographic projections;
- said top composite video display comprising a field of view of 210°;
- a multi-platform compatible processed 360° video file comprising said plurality of composite video displays; and

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said multi-platform compatible processed 360° video file viewable on the video display screen.

12. The system as recited in claim 11 wherein said plurality of video processing parameters comprises a predetermined video display layout.

13. The system as recited in claim 11 wherein said plurality of video processing parameters comprises a display height ratio.

14. The system as recited in claim 13 wherein said plurality of video processing parameters comprises a viewport field of view angle.

15. The system as recited in claim 14 wherein said plurality of video processing parameters comprises a viewport center heading angle.

16. A method for capturing, processing and displaying a 360° video on a video display screen, the method comprising:

capturing a 360° video;

generating a plurality of 360° equirectangular geographic projections;

selecting one or more of a plurality of video processing parameters, wherein the video processing parameters include a predetermined video display layout, a display height ratio, a viewport field of view angle, and a viewport heading angle;

capturing a plurality of viewport images from each of the plurality of 360° equirectangular geographic projections utilizing the one or more video processing parameters;

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generating a top composite video display comprising a plurality of viewport images corresponding to a portion of the 360° equirectangular geographic projections, said top composite video display comprising a field of view of 210°;

combining the top composite video display from each of the plurality of 360° equirectangular geographic projection to form a composite video display; and

creating a multi-platform compatible processed 360° video file comprising a plurality of composite video displays, wherein the multi-platform compatible processed 360° video file is viewable on the video display screen.

17. The method as recited in claim 16 wherein the 360° video processing assembly captures a plurality of viewport images from each of the plurality of 360° equirectangular geographic projections at a rate of about thirty frames per second to about sixty frames per second.

18. The method as recited in claim 16 further comprising capturing a plurality of geographical positioning coordinates corresponding to each of the plurality of captured viewport images, creating a geographical positioning coordinate file from the plurality of geographical positioning coordinates corresponding to each of the plurality of captured viewport images, and synchronizing each of the plurality of captured viewport images with the corresponding geographical positioning coordinates from the geographical positioning coordinate file.

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